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Grand Theme Review

Genetic and reproductive strategies to prevent mitochondrial diseases

Noemi Castelluccio [1], Katharina Spath[2], Danyang Li[3], Irenaeus F.M. De Coo[4], Lyndsey Butterworth[5], Dagan Wells[3,6], Heidi Mertes [7], Joanna Poulton[3], and Björn Heindryckx [1,*]

[1]Ghent-Fertility And Stem cell Team (G-FaST), Department for Reproductive Medicine, Department of Human Structure and Repair, Ghent University Hospital, Ghent, Belgium
[2]Juno Genetics Italy, Rome, Italy
[3]Nuffield Department of Women's and Reproductive Health, University of Oxford, Oxford, UK
[4]Department of Translational Genomics, Mental Health and Neuroscience Research Institute, Maastricht University, Maastricht, The Netherlands
[5]FutureNeuro Research Ireland Centre for Translational Brain Science, RCSI University of Medicine and Health Sciences, Dublin, Ireland
[6]Juno Genetics UK, Oxford, UK
[7]Department of Philosophy and Moral Sciences and Department of Public Health and Primary Care, Ghent University, Ghent, Belgium

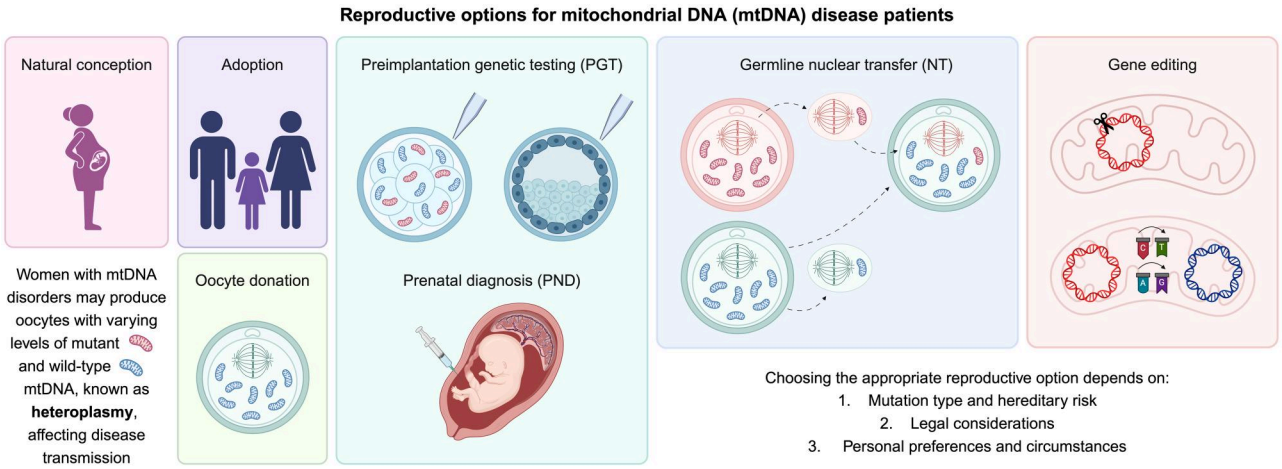
*Correspondence address. Ghent-Fertility And Stem cell Team (G-FaST), Department for Reproductive Medicine, Department of Human Structure and Repair, Ghent University Hospital, Corneel Heymanslaan 10, Entrance 37, 9000 Ghent, Belgium. E-mail: bjorn.heindryckx@ugent.be https://orcid.org/0000-0003-0630-8420

TABLE OF CONTENTS

- Introduction
- Mitochondrial genetics and function
- Mitochondrial diseases
  - Nuclear DNA (nDNA) mutations
  - Mitochondrial DNA (mtDNA) mutations
  - Diagnostic methods
  - Current treatment and long-term management of symptoms
- Impact of mitochondrial diseases on reproduction
  - Female fertility
  - Male fertility
- Genetic and reproductive options
  - Reproductive genetic testing
  - Germline nuclear transfer (NT)
  - Mitochondrial gene editing
- Emerging research and future directions
- Legal and ethical considerations
- Conclusions

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GRAPHICAL ABSTRACT



Reproductive options to prevent the transmission of mitochondrial DNA (mtDNA) diseases. Figure created with BioRender. Reproductive options to prevent the transmission of mitochondrial DNA (mtDNA) diseases include accepting the risk of transmission through natural conception or having non-genetically related children through adoption or oocyte donation. Additionally, reproductive genetic testing, including preimplantation genetic testing (PGT) and prenatal diagnosis (PND) are available. Germline nuclear transfer (NT) and gene editing have also emerged as novel techniques in this field.

ABSTRACT

Mitochondrial DNA (mtDNA) diseases pose unique challenges for genetic counselling and require tailored approaches to address recurrence risks and reproductive options. The intricate dynamics of mtDNA segregation and heteroplasmy shift significantly impact the chances of having affected children. In addition to natural pregnancy, oocyte donation, and adoption, IVF-based approaches can reduce the risk of disease transmission. Prenatal diagnosis (PND) and preimplantation genetic testing (PGT) remain the standard methods for women carrying pathogenic mtDNA mutations; nevertheless, they are not suitable for every patient. Germline nuclear transfer (NT) has emerged as a novel therapeutic strategy, while mitochondrial gene editing has increasingly become a promising research area in the field. However, challenges and safety concerns associated with all these techniques remain, highlighting the need for long-term follow-up studies, an improved understanding of disease mechanisms, and personalized approaches to diagnosis and treatment. Given the inherent risks of adverse maternal and child outcomes, careful consideration of the balance between potential benefits and drawbacks is also warranted. This review will provide critical insights, identify knowledge gaps, and underscore the importance of advancing mitochondrial disease research in reproductive health.

Keywords: mitochondrial DNA (mtDNA) / mitochondrial disease / preimplantation genetic testing (PGT) / prenatal diagnosis (PND) / germline nuclear transfer (NT) / mitochondrial gene editing

Introduction

Inherited mitochondrial diseases are complex multisystem conditions characterized by defects in oxidative phosphorylation (OXPHOS). Such disorders are often associated with a high disease burden and a significant risk of mortality. Mutations in either nuclear (nDNA) or mitochondrial DNA (mtDNA) can give rise to these conditions, reflecting the dual genetic control of mitochondrial function. The resulting genetic heterogeneity can make diagnosis difficult. Given the limited effective treatments available and the lack of known cures, couples carrying genetic mutations associated with mitochondrial diseases often seek reproductive advice to minimize the risk of having a severely affected child.

This review aims to highlight the substantial advances made in mitochondrial disease research, with a particular focus on the implications for reproduction. New opportunities to prevent the transmission of mitochondrial diseases have emerged in recent years, based on novel insights into mitochondrial genetics and function. While these procedures offer hope for the patients, they also raise important legal and ethical issues that must be carefully considered. Further research is needed to fully understand the long-term safety and efficacy of these techniques and to develop personalized approaches to diagnosis and treatment.

Mitochondrial genetics and function

Mitochondria are typically described as the powerhouses of the cell, and their primary function is to generate energy in the form of adenosine triphosphate (ATP). Additionally, they participate in numerous other cellular processes, including lipid and carbohydrate metabolism, calcium homeostasis, haeme biosynthesis, and apoptosis (Spinelli and Haigis, 2018). Structurally, mitochondria consist of outer and inner membranes, separated by an intermembrane space. The inner membrane folds into numerous cristae, extending into the organelle's interior, referred to as the matrix (Palade, 1953).

Mitochondria contain their own genetic material, known as mtDNA. In humans, this small, double-stranded, and circular chromosome comprises 16 569 base pairs (bp) and codes for 37 genes, distributed between the heavy (H) and light (L) strands. Among its encoded elements, there are 13 subunits of the electron transport chain (ETC), responsible for producing the majority of cellular ATP, as well as 22 transfer RNAs (tRNAs) and two ribosomal RNAs (rRNAs), which are involved in mitochondrial protein translation (Anderson *et al.*, 1981; Andrews *et al.*, 1999) (Fig. 1). The remaining 1136 proteins necessary for mitochondrial function and survival are encoded by nDNA, emphasizing the interplay between mitochondrial and nuclear genomes in maintaining mitochondrial homeostasis (Rath *et al.*, 2021).

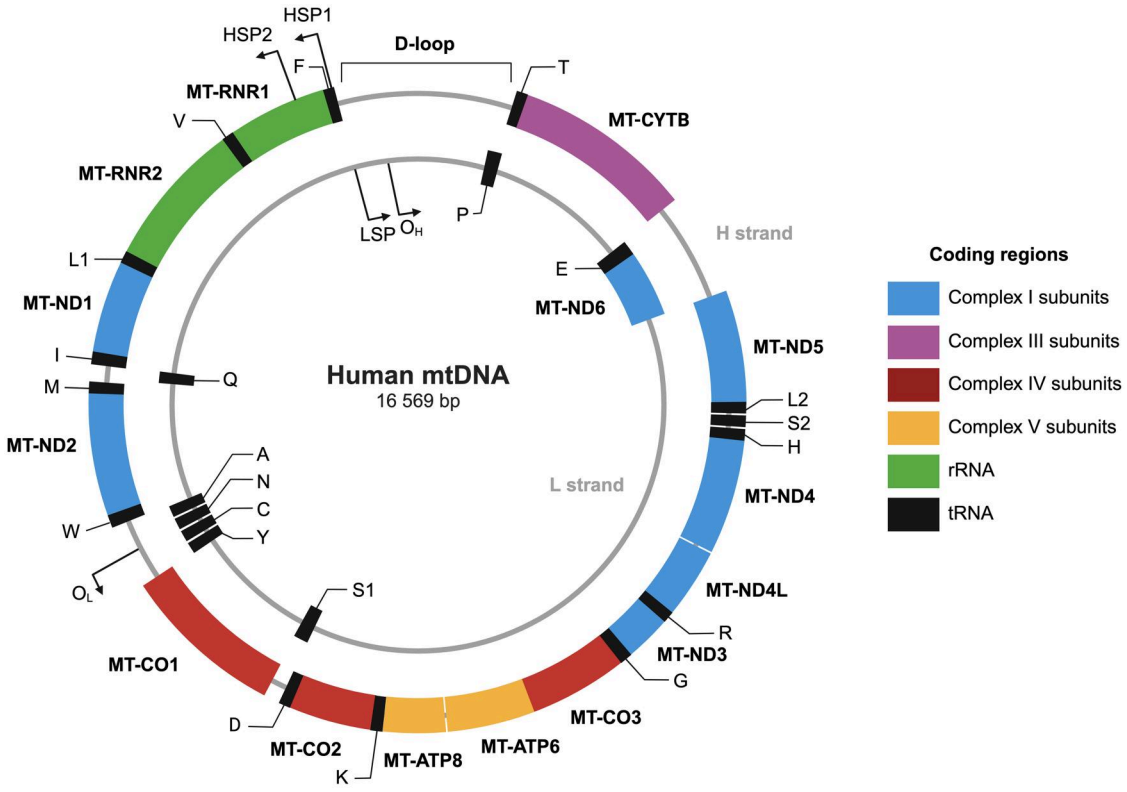


Figure 1. Mitochondrial DNA (mtDNA). The human mtDNA is a circular molecule of 16 569 bp, with an outer 'heavy' (H) strand and an inner 'light' (L) strand. The diagram highlights the 37 genes encoded by the mtDNA, including 13 essential proteins (colour-coded to indicate to which complex of the electron transport chain each subunit belongs to), as well as two ribosomal RNAs (rRNAs) (in green) and 22 transfer RNAs (tRNAs) (in black) required for their translation. Standard one-letter amino acid abbreviations are used to represent each of the tRNA molecules. There is one major non-coding region, containing the displacement loop (D-loop) (in grey). The two H-strand promoters (HSP1 and 2), the L-strand promoter (LSP), and the origin of replication for the H strand (OH) are located on the H strand, while the L strand origin of replication (OL) is downstream on the L strand. Figure created with BioRender.

Apart from having a smaller number of protein-coding genes, another notable distinction from nDNA is that ~93% of the mtDNA sequence is a coding region. Human mtDNA shows remarkable compactness and contains almost no introns, with two pairs of protein-coding regions even overlapping. The primary non-coding region of the mitochondrial genome is the displacement loop (D-loop), which serves as a control region for both transcription and replication processes within the mitochondrion. It contains the H-strand origin of replication and the major promoters for transcription, as well as three hypervariable regions (Lott *et al.*, 2013; Vallbona-Garcia *et al.*, 2023) (Fig. 1). Another function of the D-loop is to mediate mtDNA interaction with nuclear-encoded mitochondrial factors (Fisher *et al.*, 1987; Clayton, 2000).

The mitochondrial genome is intricately organized within structures known as nucleoids, each typically housing a single copy of mtDNA (Kukat *et al.*, 2015). Mitochondria form a dynamic network that adapts to the cellular metabolic needs in response to physiological and environmental signals by dividing (fission) and joining (fusion) (Scott and Youle, 2010). Fission divides a single mitochondrion into two or more independent structures, while fusion reunites separated membranes (Lewis and Lewis, 1914). These processes help to determine the shape, size, and number of mitochondria within a cell.

While mtDNA replication is more rapid during S-phase (Pica-Mattoccia and Attardi, 1972), it can continue independently of the cell nucleus through all stages of the cell cycle (Gustafsson *et al.*, 2016). Despite this autonomy, the mitochondrial replisome is encoded by the nuclear genome (Rath *et al.*, 2021; Gupta *et al.*, 2023). Key components of the replication machinery include: DNA polymerase gamma, which is responsible for mtDNA replication and repair (Ropp and Copeland, 1996); mitochondrial transcription factor A (TFAM), which is pivotal for initiating transcription and replication, as well as enhancing mtDNA packaging and compaction (Ngo *et al.*, 2014); and the helicase Twinkle, which is crucial for mtDNA copy number regulation (Tyynismaa *et al.*, 2004) and involved in catalyzing DNA unwinding and fork progression (Milenkovic *et al.*, 2013).

A key aspect of mitochondrial genetics is the fact that each cell contains multiple copies of mtDNA, ranging from a few hundred to many thousands, depending on the cell type and function (Cavelier *et al.*, 2000). This diversity is particularly evident in gametes, where mtDNA copy number typically varies from less than 100 copies in sperm to over 100 000 in a mature oocyte (Steuerwald *et al.*, 2000; Reynier *et al.*, 2001).

The term 'homoplasmy' is used to describe a state where all mtDNA copies within a cell or organism are identical, while 'heteroplasmy' refers to the presence of more than one mtDNA variant (Fig. 2). Heteroplasmy becomes significant when a pathogenic mtDNA mutation is present. In individuals with heteroplasmy, disease expression is determined by the proportion of mutated mtDNA genomes relative to the total number of mtDNA molecules, by convention expressed as a percentage. In many heteroplasmic disorders, the greater the load of mutant mtDNA, the more likely a cell is to exhibit a phenotype. Disease manifestation occurs only when a critical threshold level is exceeded,

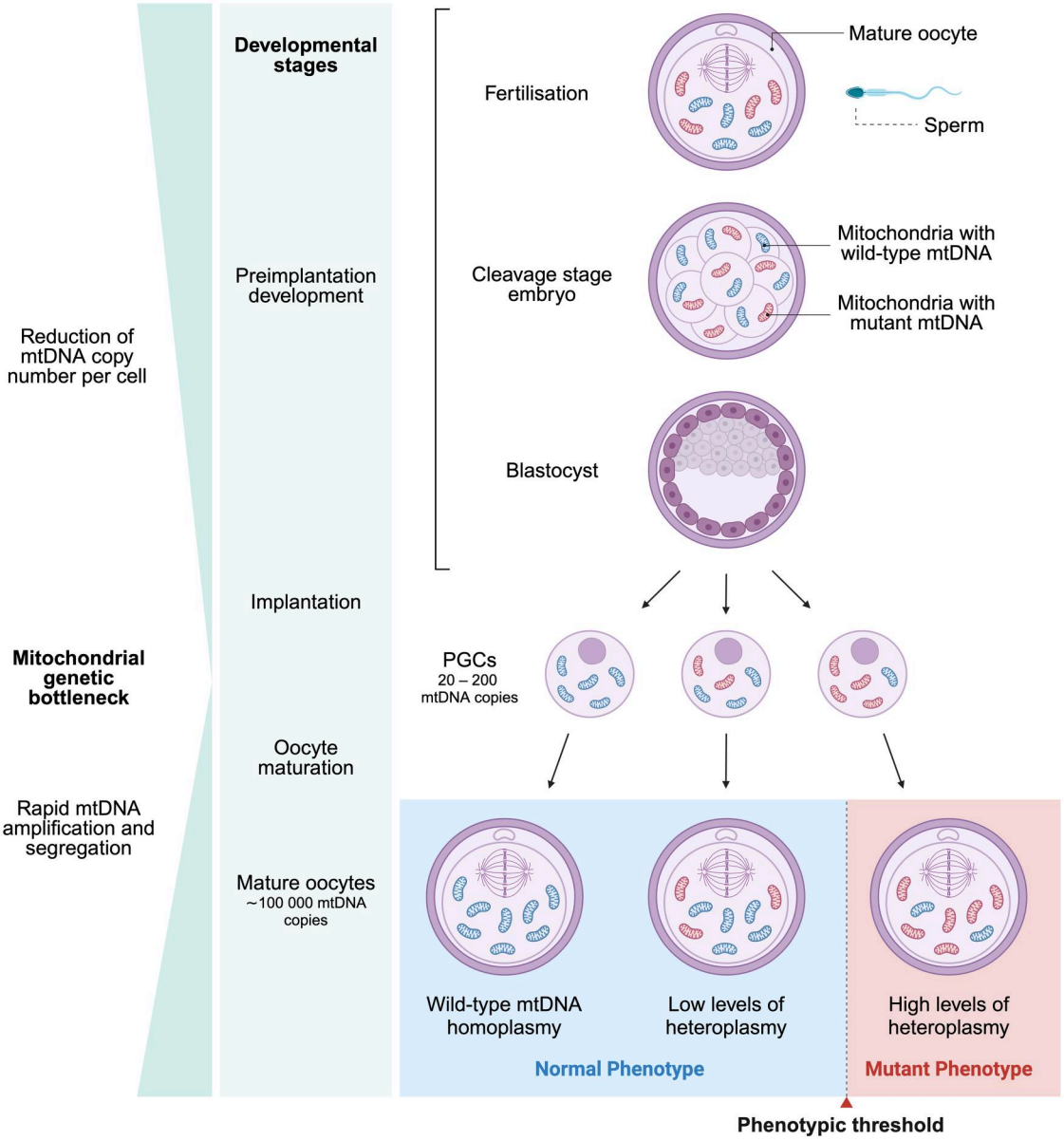


Figure 2. The mitochondrial genetic bottleneck and heteroplasmy. The reduction of mtDNA copies from a broad diversity in primordial germ cells (PGCs) through a bottleneck during oogenesis (20–200 mtDNA copies in PGCs) results in a large variance of mtDNA proportion in the mature oocyte (~100 000 mtDNA copies in mature cells). The presence of both wild-type (blue) and mutant (red) mtDNA illustrates heteroplasmy, with the potential to maintain a normal phenotype or develop a mutant phenotype depending on whether the proportion of mutant mtDNA crosses a critical threshold. Figure created with BioRender.

known as the threshold effect (Rossignol *et al.*, 2003). This often depends on the specific mutation and the tissue type (Shoffner *et al.*, 1990; Boulet *et al.*, 1992), ranging from nearly 100% to as low as 8% with a rare anticodon mutation (Sacconi *et al.*, 2008). The degree of heteroplasmy may vary within cells of the same tissue or organ, between organs of the same individual, and among individuals of the same family (Rahman *et al.*, 2001; De Laat *et al.*, 2012).

Another distinctive feature of mtDNA is its maternal inheritance pattern (Giles *et al.*, 1980; Pyle *et al.*, 2015), with paternal mtDNA being degraded post-fertilization (Kaneda *et al.*, 1995), likely through a ubiquitination mechanism (Sutovsky *et al.*, 1999). Studies in mice have demonstrated that an admixture of normal yet distinct mouse mtDNAs can result in genetic instability and adverse physiological effects, which might explain the evolutionary advantage of uniparental inheritance of mtDNA (Sharpley *et al.*, 2012). Paternal inheritance has, however, been observed in some exceptionally rare cases (Luo *et al.*, 2018). Maternal inheritance poses a substantial risk for women carrying a pathogenic mtDNA mutation, which is further complicated by another unique characteristic of mtDNA known as the mitochondrial genetic bottleneck.

During early embryonic development, there is a significant reduction in mtDNA copy number until the specification of primordial germ cells (PGCs), which are characterized by a minimal number of founding mitochondria. This is followed by a marked amplification process until the mature metaphase II (MII) oocyte stage. Biologically, the bottleneck may function to eliminate highly deleterious mutations by preventing the development of germ cells with elevated heteroplasmic levels of dysfunctional mtDNA. For example, evidence from mouse studies demonstrates strong purifying selection against nonsynonymous

mutations in protein-coding genes in the female germline, whereas rRNA and tRNA genes undergo less intense purifying selection (Stewart *et al.*, 2008). In addition, the bottleneck can cause marked variation in heteroplasmy levels between and within generations, as all mtDNA copies present in the mature oocyte will have originated from a very small initial pool (Cree *et al.*, 2008; Wai *et al.*, 2008; Neupane *et al.*, 2015) (Fig. 2). Such variability complicates the ability to predict the mitochondrial genotype of an individual oocyte, making genetic counselling for women with pathogenic mtDNA mutations particularly challenging. Asymptomatic carriers with a low proportion of mutant mtDNA in somatic cells may produce oocytes with very high mutation loads, potentially resulting in the manifestation of the disease in their children. Implications of the genetic bottleneck and heteroplasmy for mtDNA disorders will be discussed in more detail in the *Heteroplasmy and the mitochondrial bottleneck* section.

Replication of mtDNA is active throughout oogenesis but halts in the mature oocyte, with early embryonic development relying almost entirely on the mitochondrial pool of the oocyte. However, a study by McConnell and Petrie (2004) revealed a brief window after fertilization where mtDNA synthesis occurs, influenced by environmental factors, such as embryo culture conditions. Subsequently, the mitochondrial copy number gradually decreases in individual cells of the cleavage-stage embryo as the mitochondrial content of the oocyte is distributed to the different blastomeres. Limited human data suggest that the embryonic mtDNA content remains stable during early development, with the eight-cell stage embryo displaying comparable mtDNA copy numbers to an unfertilized oocyte (Steuerwald *et al.*, 2000; Lin *et al.*, 2004; Hashimoto *et al.*, 2017). Mitochondrial biogenesis resumes progressively, coinciding with the beginning of cell differentiation. At the blastocyst stage, increased protein synthesis and blastocoel expansion correspond to increased mitochondrial activity in trophectoderm (TE) cells that gives rise to extraembryonic tissues. In contrast, activation of mitochondrial function in the inner cell mass (ICM), forming the embryo proper, is potentially delayed until the post-implantation stages (Houghton, 2006; May-Panloup *et al.*, 2021). However, the precise timing and mechanism remain unknown.

Mitochondrial diseases

Mitochondrial diseases are caused by mutations in either the nuclear or the mitochondrial genomes. Studies reveal that pathogenic mtDNA mutations are more common than previously understood, with about 1 in every 200 individuals carrying such mutations, and of these, about half carry the m.3243A>G (Manwaring *et al.*, 2007; Elliott *et al.*, 2008) mutation. However, the prevalence of adults presenting with actual mitochondrial disease is about 1 in 5000 for mtDNA mutations, while the total prevalence is ~1 in 4300 when including both mtDNA and nDNA mutations (Gorman *et al.*, 2015b).

Nuclear DNA (nDNA) mutations

As nDNA encodes numerous proteins essential for mitochondrial structure and function, mutations in nuclear genes can lead to mitochondrial disease. These disorders, often inherited in autosomal recessive or dominant patterns, reflect the central role of mitochondria in energy production, apoptosis, and cellular signalling.

The most common group of mitochondrial disorders involving nuclear genes are those affecting mtDNA maintenance, caused by abnormal nucleotide homeostasis or defects of the mitochondrial replisome. A prime example of this mechanism is seen in mutations in the *POLG* gene, which encodes the catalytic subunit of DNA polymerase gamma, essential for replicating and repairing mtDNA (Kujoth *et al.*, 2005). Hence, *POLG* mutations result in insufficient mtDNA synthesis (El-Hattab and Scaglia, 2013). Disorders resulting from *POLG* mutations are classified based on age of onset and range from adult-onset myopathy, known as chronic progressive external ophthalmoplegia (CPEO), to fatal early-onset Alpers-Huttenlocher syndrome due to liver dysfunction and neurological involvement (Horvath *et al.*, 2006; Hikmat *et al.*, 2020). Alpers syndrome is the most common childhood-onset mtDNA depletion syndrome. Mutations in genes such as deoxyguanosine kinase (*DGUOK*) and mitochondrial thymidine kinase 2 (*TK2*) disrupt nucleotide homeostasis, affecting the biogenesis of mitochondrial deoxyguanosine triphosphate (dGTP) and deoxythymidine triphosphate (dTTP), respectively (Mandel *et al.*, 2001; Saada *et al.*, 2001).

Pathogenic mutations have also been identified in genes such as mitochondrial inner membrane protein MPV17 (*MPV17*) (Spinazzola *et al.*, 2006), OPA1 mitochondrial dynamin-like GTPase (*OPA1*) (Alexander *et al.*, 2000), mitofusin 2 (*MFN2*) (Züchner *et al.*, 2004), SURF1 cytochrome c oxidase assembly factor (*SURF1*) (Tiranti *et al.*, 1998; Zhu *et al.*, 1998), PTEN-induced kinase 1 (*PINK1*) (Valente *et al.*, 2004), and parkin RBR E3 ubiquitin protein ligase (*PRKN*) (Shimura *et al.*, 2000), all of which are involved in mitochondrial fusion, dynamics and ETC assembly. Transcription of mtDNA is also a critical process that is essential for gene expression and the regulation of mtDNA copy number, but defects are less commonly diagnosed.

Mitochondrial DNA (mtDNA) mutations

While nDNA mutations play a significant role in mitochondrial diseases, specific mutations within the mitochondrial genome also contribute substantially.

Homoplasmy versus heteroplasmy

Homoplasmy and heteroplasmy play a crucial role in the phenotype and severity of mitochondrial diseases. Notable homoplasmic mtDNA mutations affect the mitochondrially encoded NADH: ubiquinone oxidoreductase core subunits, including m.11778G>A in *MT-ND4* (Wallace *et al.*, 1988), m.14484T>C in *MT-ND6* (Johns *et al.*, 1992), and m.3460G>A in *MT-ND1* (Howell *et al.*, 1995), all associated with Leber's hereditary optic neuropathy (LHON); the m.1555A>G mutation in the mitochondrially encoded 12S rRNA (*MT-RNR1*) gene linked to mitochondrial non-syndromic sensorineural hearing loss (Torroni *et al.*, 1999); and the m.4300A>G mutation in the mitochondrially encoded tRNA-Ile (*MT-TI*) gene, associated with mitochondrial hypertrophic cardiomyopathy (Taylor *et al.*, 2003).

Homoplasmic mtDNA disease is variable, even though all individuals share the same genotype, as disease penetrance is influenced by both genetic and environmental factors, including lifestyle choices such as smoking and variation in mtDNA copy number. For example, smoking has been identified as a major environmental risk factor that exacerbates mitochondrial dysfunction, increasing the likelihood of disease manifestation among mutation carriers (Kirkman *et al.*, 2009). Similarly, individuals with higher mtDNA copy numbers may exhibit some resilience against the phenotypic expression of LHON (Korsten *et al.*, 2010).

Conversely, heteroplasmic mutations, such as m.3243A>G in the mitochondrially encoded tRNA-Leu 1 (*MT-TL1*) gene, present a different clinical picture. Higher heteroplasmy levels in metabolically active tissues can lead to more severe manifestations of mitochondrial encephalomyopathy with lactic acidosis and stroke-like episodes (MELAS) syndrome, while lower levels may

result in milder or subclinical presentations. Mutant mtDNA levels fluctuate over time, decreasing in blood (Rahman et al., 2001) and increasing in muscle (Weber et al., 1997). However, a single measurement of mutant load has very poor predictive value, even in muscle tissue (Grady et al., 2018). The severity and progression of associated diseases are linked to age-adjusted heteroplasmy levels (Grady et al., 2018), yet this relationship can be nuanced by factors such as tissue-specific expression, compensatory mechanisms within cells, and external influences including environmental exposures. Additionally, nuclear genetic modifiers and the mitochondrial-nuclear genomic interaction play significant roles in disease expression, further complicating the predictive value of heteroplasmy alone (Pickett et al., 2018). Sex differences in disease phenotype and progression have also been observed, suggesting that chromosomal sex might influence the impact of the m.3243A>G mutation (Pickett et al., 2018).

Structural rearrangements in mtDNA, including large-scale deletions and duplications, are also heteroplasmic. The 'common deletion' (m.8483_13459del4977) includes 4977 bp harbouring both protein reading frames and tRNA genes. Most mtDNA duplications co-exist with related rearrangements, including partial deletions and/or deletion dimers (Poulton et al., 1993), but they can occur in isolation. Single large-scale mtDNA deletions commonly present as ocular myopathies, with one example being Kearns-Sayre syndrome (KSS), characterized by endocrinopathies, deafness, and progressive external ophthalmoplegia (PEO). In early life, these deletions may manifest as Pearson's syndrome with sideroblastic anaemia in infancy that spontaneously resolves as the rearranged mtDNAs in blood decline, followed by liver dysfunction and pancreatic insufficiency. If the children survive, they then develop the adult KSS phenotype (McShane et al., 1991), including muscle weakness with PEO, retinopathy, endocrinopathy, and cardiac conduction defects with central nervous system involvement. However, clinical features are often heterogeneous, with phenotypes falling within a spectrum of diseases with multisystem complications (Mancuso et al., 2015; Björkman et al., 2023).

A systematic review of 159 heteroplasmic mtDNA mutations spotlighted three prevalent mutations (m.3243A>G, m.8344A>G, m.8993T>C/G), with mutation levels having some predictive value. Significantly, muscle mutation levels under 18% correlated with a greater than 95% probability of not manifesting related disorders, guiding reproductive decisions for families with mtDNA mutations (Hellebrekers et al., 2012).

Heteroplasmy and the mitochondrial bottleneck

Heteroplasmy is central to the variability and progression of mtDNA diseases and underlies the mitochondrial bottleneck, a unique phenomenon in mtDNA transmission. This was first observed by Ashley et al. (1989) who documented complete switching of mtDNA type in a single generation in Holstein cows. The bottleneck implies that a small number, or even a single mtDNA, may become the founder for offspring (Fig. 2). This discovery, which is also termed segregation (Burgstaller et al., 2014), has been substantiated in various contexts, including mitosis, oocytes, and mouse models.

Marchington et al. (1998) showed that single mouse or human oocytes from the same mother were populated by different mtDNA clones, inferring that a major component of the bottleneck occurs by the time oocytes are mature (Marchington et al., 1997). This is consistent with the substantial increase in mtDNA copy number between PGCs and mature oocytes. Studies by Jenuth et al. (1996) and Zhang et al. (2021) revealed early segregation of mtDNA variants between PGCs and primary oocytes, shifting our understanding of mtDNA dynamics during development (Fig. 2). The use of mathematical models, like the Kimura distribution, has delineated the random patterns of mtDNA segregation in inheritance (Wonnapinij et al., 2008). However, this field remains rife with questions about potential selection biases or skewed distributions.

Intriguingly, Battersby et al. (2003) performed a linkage analysis, but the genes identified provided little insight into the mechanisms regulating segregation. Burgstaller et al. (2018) quantitatively demonstrated that heteroplasmy variance is influenced by mtDNA composition and tissue type, with segregation in oocytes correlating with maternal age, consistent with mtDNA selection. This selection could be at the level of the whole cell ('fitness') or reflect mtDNA turnover. Selection within the mitochondrion might involve differences in replication rates (Emmerson et al., 2001) or in mitophagy, a selective autophagic process eliminating defective mitochondria (Kiššová et al., 2004; Lemasters, 2005). Although evidence supporting mitophagy in germline segregation is limited (Diot et al., 2016; Glynos et al., 2023; Kremer et al., 2023), the concept of mtDNA 'selfishness' (e.g. the balance between replication and transcription) and environmental factors, like oxygen tension, have advanced our understanding of mtDNA segregation (Emmerson et al., 2001; Pezet et al., 2021).

At the cellular level, the pattern of mtDNA segregation in human oocytes differs significantly between pathogenic mutations in tRNA genes (Marchington et al., 1998; Brown et al., 2001) and those in protein-coding genes like m.8993T>G in the mitochondrially encoded ATP synthase membrane subunit 6 (*MT-ATP6*) gene (Blok et al., 1997). The hyperpolarized mitochondrial membrane potential in cells with high m.8993T>G mutation loads might drive mitochondria into Balbiani bodies, embryonic structures thought to play a role in mtDNA transmission (Otten et al., 2018).

Inheritance of heteroplasmy

The development and progression of mitochondrial diseases are deeply influenced by the concept of heteroplasmy shift, which refers to the changing proportions of mtDNA populations or mutant mtDNA load within cells and generations. This shift is shaped by various factors such as random genetic drift, replicative advantage of specific mtDNA variants, and mitophagy. Monnot et al. (2011) highlighted differences in bottleneck size, a concept confirmed by Wilson et al. (2016). Further complexities in heteroplasmy dynamics include vegetative segregation, which increases variance with ongoing cell divisions (Preiss et al., 1995), and relaxed replication, where mtDNA replication continues in non-dividing cells, impacting mtDNA distribution among daughter cells (Stewart and Chinnery, 2015). Notably, postmitotic cells, like skeletal muscle, show more stable and higher heteroplasmy levels over time due to purifying selection, unlike rapidly dividing cells where levels of mutant mtDNA tend to fall (Rahman et al., 2001).

Programmed mitophagy plays a key role when Drosophila germ cells enter meiosis, driven by mitochondrial fragmentation (Lieber et al., 2019), although this does not appear to depend on mtDNA or improve mitochondrial quality (Palozzi and Hurd, 2023). Despite ongoing research into mtDNA heteroplasmy in the mammalian germline, evidence for the mechanisms behind these changes or the occurrence of mitophagy remains sparse (Diot et al., 2016; Glynos et al., 2023; Kremer et al., 2023). Wai et al. (2008) used a heteroplasmic mouse model to demonstrate both heteroplasmic shift and mtDNA synthesis with a 700-fold

expansion in mtDNA copy number between PGCs and mature oocytes. Glynos *et al.* (2023) confirmed these findings, showing that this coincided with a clonal expansion of certain mtDNA heteroplasmies, driven by changes in metabolism and gene expression. They suggested that a metabolic shift might exert selection pressure and confer a disadvantage on some clones carrying deleterious mutations. Similar heteroplasmy was detected in early human embryos, although the study was limited to early germline development (Floros *et al.*, 2018). It is now clear that mtDNA point mutations can arise at any stage of embryogenesis and human development (Mertens *et al.*, 2022) and may be increased by ART (Mertens *et al.*, 2024), but with unknown consequences. Floros *et al.* (2018) also confirmed that the human genetic bottleneck is narrower than that of the mouse, as earlier data suggested (Poulton *et al.*, 1998). Using the POLG 'mutator' mouse strain that harbours many heterogeneous mtDNA point mutations, Ma *et al.* (2020) demonstrated purifying selection from oocyte through embryogenesis, pointing to an effective post-implantation selection mechanism. Moreover, Klucnika and Ma (2019) investigated the transmission of mtDNA and the conflict between purifying selection opposing selfish mtDNA strains in flies. They revealed that reducing mtDNA polymerase POLG restricts the transmission of mtDNA mutations (Chiang *et al.*, 2019), likely by affecting mtDNA replication in particular genetic backgrounds (Ma *et al.*, 2014).

The investigation into mtDNA segregation and heteroplasmy shift is pivotal for understanding mitochondrial diseases. Research reveals that these dynamics, shaped by genetic bottlenecks, selective pressures, and quality control mechanisms like mitophagy, are essential in determining disease outcomes.

Diagnostic methods

Patients with mitochondrial disease, with their often complex and multifaceted symptomatology, require a team of specialists for diagnosis, treatment, preventive care, and coordinated multidisciplinary support. To improve the assessment of patients suspected of mitochondrial disease, mitochondrial disease criteria for children (Wolf and Smeitink, 2002; Morava *et al.*, 2006; Witters *et al.*, 2018) and adults (Bernier *et al.*, 2002) have been developed. Accurate and timely diagnosis of mitochondrial disease is crucial for effective management and treatment.

A diagnostic workup starts with a thorough clinical examination, preferably by healthcare professionals experienced in mitochondrial disorders. Symptoms such as muscle weakness, fatigue, exercise intolerance, and organ dysfunction are frequently encountered and may raise suspicion of mitochondrial disease. However, the spectrum of symptoms is much broader, and the overlap with other conditions poses challenges.

Further assistance to make a diagnosis has been made via publishing consensus statements (Emmanuele *et al.*, 2022). Their focus has moved from discovering patients with mitochondrial disease to criteria to standardize diagnosis and categorization of mitochondrial disease to document natural history studies and clinical trials (Phoenix *et al.*, 2006; Schaefer *et al.*, 2006; Koene, 2017).

Except for some delineated clinical mitochondrial syndromes (e.g. Leigh, MELAS, CPEO) (Emmanuele *et al.*, 2022), diagnosing mitochondrial disease requires a comprehensive laboratory approach. This typically includes biochemical tests, molecular genetic analysis (both nuclear and mitochondrial), neuroimaging, pathology (of skeletal muscle and liver), and functional assays on tissues (Parikh *et al.*, 2015).

Genetic testing helps in identifying disease-causing mutations and assessing familial inheritance patterns. Next-generation sequencing (NGS) has enhanced diagnostic accuracy, achieving yields between 35% and 70% for mitochondrial disorders (Theunissen *et al.*, 2018; Stenton and Prokisch, 2020). In cases where no (likely) pathogenic variants have been identified, it remains important to continue the diagnostic search, because a clear diagnosis is crucial for accurate prognosis, effective treatment planning, and informed genetic counselling (Ng *et al.*, 2021; Wortmann *et al.*, 2022).

Biochemical assays are used to evaluate mitochondrial function and metabolism. Blood, urine, and tissue samples are analysed for abnormalities in biomarkers such as lactate, pyruvate, amino acids, and respiratory chain enzyme activities. However, while these methods provide valuable insights, they may lack sensitivity and specificity, necessitating additional confirmatory testing or use of them in combination. For instance, a panel including lactate, acylcarnitine profiling, growth differentiation factor-15 (GDF-15), fibroblast growth factor-21 (FGF-21), and mtDNA copy number might cover the heterogeneity of these multifactorial diseases (Hubens *et al.*, 2022).

Imaging modalities such as MRI, proton magnetic resonance spectroscopy (1H-MRS), and computed tomography (CT) complement genetic and biochemical findings. They support diagnosis by visualizing structural abnormalities or metabolism in affected organs, such as the brain, heart, and muscles (Alves and Whitehead, 2024).

Pathology is a valuable tool, especially when biochemical analysis and genetic testing are inconclusive. Histological analysis of muscle tissue can reveal characteristic abnormalities, such as ragged-red fibres and mitochondrial cytochrome c oxidase-negative fibres. Moreover, electron microscopy allows for a detailed examination of mitochondrial ultrastructure (Parikh *et al.*, 2015).

Functional assays that directly assess mitochondrial function, such as high-resolution respirometry (e.g. Seahorse XF Analyzer) for mitochondrial respiration, mass spectrometry-based ATP quantification, and fluorescence or chemiluminescence-based methods for detecting reactive oxygen species (ROS), are increasingly used in patient-derived cells or tissues. These assays provide insights into mitochondrial dysfunction at the cellular level and may aid in personalized treatment strategies.

Ongoing research aims to identify novel biomarkers for early diagnosis and monitoring of mitochondrial diseases. Potential biomarkers, including circulating mtDNA, metabolites, and mitochondrial-derived peptides, show promise in reflecting mitochondrial dysfunction and disease severity. Large-scale studies are needed to validate these biomarkers for clinical use (Hubens *et al.*, 2022).

Current treatment and long-term management of symptoms

The chronic nature of many mitochondrial diseases requires lifelong support. General recommendations for mitochondrial disease care management have been published in consensus papers and guidelines (Parikh *et al.*, 2015, 2017; Sue *et al.*, 2022). Specific attention has been given to the management of strokes (Ng *et al.*, 2019), seizures (Mancuso *et al.*, 2024), drug safety (De Vries *et al.*, 2020), nutrition (DiVito *et al.*, 2023), and reproductive options (Turnbull *et al.*, 2013; Poulton *et al.*, 2019; Smeets *et al.*, 2023). Recently, the psychosocial and psychiatric aspects of mitochondrial disorders have also gained more attention, with shared decision-making gradually making its way into the clinic (Moore *et al.*, 2023).

While there is currently no curative treatment for mitochondrial diseases, significant progress has been made in managing symptoms and improving the quality of life for affected

individuals. Symptomatic treatment of specific organ systems affected by mitochondrial dysfunction, such as muscle weakness, cardiac abnormalities, neurological deficits, and metabolic disturbances, can be targeted by pharmacological interventions, physical therapy, occupational therapy and nutritional support tailored to the individual patient's needs.

Supplementation with cofactors and antioxidants aims to optimize mitochondrial function and alleviate oxidative stress. These approaches are reviewed in more detail elsewhere (Zong et al., 2024), but a brief summary is provided here. For instance, coenzyme Q10 (CoQ10) is used for primary COQ8A-ataxia (Traschütz et al., 2020), and Idebenone for LHON (Carelli et al., 2017). Patients with disorders of vitamin cofactor metabolism (e.g. thiamine, biotin, riboflavin) and multiple acyl-CoA dehydrogenase deficiency might benefit from specific supplements (Distelmaier et al., 2017), while deoxynucleoside treatment has shown positive results in TK2 deficiency (Domínguez-González et al., 2019).

Recent advances in gene and cell therapy offer exciting prospects for treating mitochondrial disorders by targeting underlying genetic defects. For example, an adeno-associated virus (AAV)-based gene therapy is currently in phase 3 clinical trial for LHON (Vignal-Clermont et al., 2023). In addition, cell therapy provides the possibility to regenerate damaged or dysfunctional tissues of mitochondrial disease patients. Systemic administration of healthy autologous myogenic stem cells to combat myopathy and fatigue in m.3243A>G patients is currently in phase 2a clinical trial (clinicaltrials.gov—study NCT05962333) (Van Tienen et al., 2019).

These developments in treatment modalities, combined with a holistic approach that integrates symptomatic management, mitochondrial supplements, exercise therapy, and advanced therapies, give hope for improved outcomes and enhanced quality of life. Addressing the multifaceted nature of mitochondrial diseases through comprehensive patient care strategies remains essential.

Impact of mitochondrial diseases on reproduction

Female fertility

Female carriers of mtDNA mutations generally display live birth rates comparable to the general population, suggesting normal fertility (Gorman et al., 2015a). However, pregnancy might present a challenge due to the heterogeneity of clinical symptoms. Existing evidence, mostly from case reports and studies with small sample sizes, limits the validity and reliability of available information. Nevertheless, care guidelines have been developed to manage pregnancy in women with mitochondrial disease (Turnbull et al., 2013; Poulton et al., 2019).

Increased energy demand during pregnancy creates a potential window for the manifestation of mitochondrial disease. While some mtDNA patients experience asymptomatic pregnancies, others may develop mild to severe symptoms, such as exercise intolerance or persistent paraesthesia (Say et al., 2011). Pregnancies in women with specific mtDNA mutations, such as m.3243A>G, show an increased risk of complications, including gestational diabetes, breathing difficulties, and hypertension, often leading to emergency caesarean sections (24.2% of cases), preterm deliveries (53.3% of cases) (<37 weeks), and neonatal complications (Nakamura et al., 2016; Feeney et al., 2019).

Additionally, for individuals undergoing IVF treatment, it is essential to be sufficiently fit to endure ovarian stimulation and pregnancy. Studies on carriers of pathogenic mtDNA mutations have revealed mixed results regarding ovarian reserve and response to ovarian stimulation. One study by Kang et al. (2016) found significantly lower levels of anti-Müllerian hormone (1.1 ng/ml compared to 4.8 ng/ml in healthy donors) and a reduced antral follicle count (10.3 compared to 22.3 in healthy donors), suggesting diminished ovarian reserve in these patients. In contrast, another study reported no significant differences in these parameters (Mayeur et al., 2023). Furthermore, carriers of mtDNA mutations generally require longer ovarian stimulation (an average of one extra day) and higher doses of gonadotropins (Kang et al., 2016; Mayeur et al., 2023). Findings regarding oocyte retrieval and maturation rates have also been inconsistent. Mayeur et al. (2023) showed no significant differences, while Kang et al. (2016) reported lower retrieval rates in mtDNA mutation carriers. These conflicting results may be attributed to the small sample sizes of these studies (n = 4 and n = 18, respectively), which limits the generalizability of the findings (Kang et al., 2016; Mayeur et al., 2023). Patients included in these reports were carriers of mutations such as m.3243A>G, m.8344A>G, m.8993T>C, and m.10197G>A.

Despite the essential role of mitochondria in oogenesis and preimplantation development, no consistent correlation has been established between pathogenic mtDNA mutations and fertilization failure, impaired oocyte quality, or embryo developmental arrest (Steffann et al., 2015). Pathogenic mtDNA mutations do not appear to affect preimplantation embryo quality or viability (Chatzovoulou et al., 2021). However, specific mitochondrial variants may downregulate gene expression in embryos, potentially impacting OXPHOS and cell survival pathways (Chatzovoulou et al., 2023).

Male fertility

In contrast to women, male carriers of mtDNA disorders exhibit a 35% decrease in fertility compared to the general population (Martikainen et al., 2017). The main role of sperm mitochondria is to generate the energy for flagellar beating. High levels of mutant mtDNA impair OXPHOS and correlate with low sperm motility, especially in carriers of the m.3243A>G mutation (Huang et al., 1994; Spiropoulos et al., 2002). Patients with major mtDNA rearrangements and point mutations often have dysfunctional mitochondria and insufficient energy production, leading to poorer sperm parameters and overall fertility (Folgerø et al., 1993; Kao et al., 1998; Ambulkar et al., 2016). Moreover, men with oligoasthenoteratozoospermia (OAT) typically carry higher numbers of mtDNA deletions in their sperm (Lestienne et al., 1997; Kao et al., 1998; St John et al., 2001). For example, large-scale rearrangements, such as the 'common deletion' (4977 bp), encompass both protein-coding genes and tRNA genes, possibly resulting in multiple respiratory chain defects (Kao et al., 1998). Specific mutations in ATP synthesis genes, such as the 11696G>A variant in the *MT-ND4* gene, have also been implicated in impaired sperm motility (Ji et al., 2017). Additionally, mtDNA mutations can lead to a significant loss of energy production in Sertoli and other supporting cells, affecting spermatogenesis and contributing to male infertility (Cummins et al., 1994).

Animal studies support these findings; transgenic mice with pathogenic mtDNA mutations show decreased sperm count, motility, and testis size, highlighting the impact of mitochondrial dysfunction on male fertility (Trifunovic et al., 2004; Nakada et al., 2006).

Genetic and reproductive options

To avoid the transmission of mitochondrial disease, couples at risk have several options available to them. The choice often

depends on the underlying mutation and the associated recurrence risk. They may choose approaches that exclude them from genetic parenthood, such as adoption or oocyte donation (Smeets *et al.*, 2023). This may be the preferred choice in women who carry homoplasmic mtDNA mutations and therefore pass the mutation on to all their children, or in couples where the genetic cause is unknown. Yet, owing to technological advances in sequencing methods, the underlying genetic cause can now be successfully determined in the great majority of couples. This gives them the possibility to opt for reproductive genetic testing, providing an opportunity to conceive a genetically related child while significantly reducing the risk of disease transmission. Reproductive genetic testing can be divided into two main categories: preimplantation genetic testing (PGT), which involves testing of embryos generated *in vitro* before transfer to the uterus, and prenatal diagnosis (PND), in which the foetus is tested after implantation (Mavraki *et al.*, 2023; Smeets *et al.*, 2023).

While PGT and PND are almost always available for couples with nDNA mutations, these tests are sometimes less useful for carriers of mtDNA mutations, especially when the mother is a homoplasmic carrier or has high mutation levels (Smeets *et al.*, 2023). In these cases, a possible option could be germline nuclear transfer (NT), a relatively new procedure in which the genetic material of the mother's oocyte or zygote is transferred into the cytoplasm of an enucleated oocyte or zygote from a healthy donor who is not a carrier of a mtDNA mutation (Craven *et al.*, 2017; Greenfield *et al.*, 2017). In addition, mitochondrial gene editing techniques have also emerged as a novel research area with possible future therapeutic applications (Moraes, 2024).

Reproductive genetic testing

Reproductive genetic testing during the preimplantation and/or prenatal stage helps reduce the risk of hereditary mitochondrial disease, provided the genetic cause is known. For both PGT and PND, the genetic analysis approach varies based on whether the mutation is in the nuclear or the mitochondrial genome. For autosomal and X-linked mutations that follow Mendelian inheritance, conventional tests are utilized (Mavraki *et al.*, 2023; Smeets *et al.*, 2023), and their application and interpretation of results are generally straightforward and therefore will not be discussed further.

In contrast, the genetic analysis of mtDNA mutations requires a qualitative and quantitative component to determine not only the presence of the mutation but also its quantity, specifically the mutation load (Mavraki *et al.*, 2023; Smeets *et al.*, 2023).

Preimplantation genetic testing (PGT)

PGT necessitates ART to create embryos *in vitro*, which undergo genetic analysis before uterine transfer. This process requires hormonally controlled ovarian stimulation, oocyte retrieval, fertilization, and embryo biopsy to assess mtDNA mutations and their levels (Fig. 3A). Suitable embryos (those free of harmful mutations or with low mutation levels) are then transferred to the uterus (Bredenoord *et al.*, 2008a; Poulton and Bredenoord, 2010).

PGT for mtDNA mutations (PGT-MT) is not a routine clinical test but a highly specialized and relatively rare procedure. Due to the complexities of mtDNA mutation inheritance, PGT-MT should only be performed in specialized centres with ample experience in both mtDNA genetics and preimplantation embryo diagnostics (Poulton and Bredenoord, 2010; Mavraki *et al.*, 2023). The close cooperation of a fertility clinic with expertise in embryo biopsy and excellent IVF success rates is also vital for an effective PGT programme. The clinical application of PGT-MT has been reported by a few centres worldwide, with only a handful of children born through this method (Thorburn *et al.*, 2009; Gigarel *et al.*, 2011; Monnot *et al.*, 2011; Vandewoestyne *et al.*, 2011; Treff *et al.*, 2012; Vandewoestyne *et al.*, 2012; Sallevelt *et al.*, 2013; Heindryckx *et al.*, 2014; Steffann *et al.*, 2014; Sallevelt *et al.*, 2015; Greenfield *et al.*, 2017; Sallevelt *et al.*, 2017b,c; Spath *et al.*, 2021) (Table 1).

PGT-MT can only minimize the risk of disease transmission but cannot eliminate this possibility. Its success relies on the biopsy result, which must accurately reflect the genetic status of the embryo. Moreover, if a mutation-free embryo is unavailable and a carrier embryo is considered for transfer, it is important that genotype–phenotype correlations are well understood, allowing determination of the mutation levels below which disease is unlikely to occur. Finally, the mutation load measured in embryo specimens must remain stable during both pre- and postnatal development to prevent later disease development (Poulton and Bredenoord, 2010). The fulfilment of these requirements is largely dependent on the specific mutation. However, since global experience with PGT-MT is very limited, the predictive value of PGT-MT can only be estimated and cannot be generalized. Consequently, PGT-MT should be considered as a risk-reduction option and not as a guarantee of an unaffected pregnancy/child (Mavraki *et al.*, 2023; Smeets *et al.*, 2023).

Challenges of sampling and analysis: timing of embryo biopsy

Theoretically, PGT-MT can be performed at three time points during preimplantation development: first, at the MII oocyte stage and/or the zygote stage by analysing the first and/or second polar body; second, at the cleavage stage by analysing one or two blastomeres; and third, at the blastocyst stage by analysing TE cells (Poulton and Bredenoord, 2010) (Fig. 3A).

Polar body biopsy may be preferred for legal or ethical reasons, or due to its less invasive nature (Gigarel *et al.*, 2011; Vandewoestyne *et al.*, 2011, 2012). However, limited data exist on the accuracy of polar body analysis, with studies in mice and humans showing conflicting results regarding the agreement between mtDNA mutation levels in polar bodies and corresponding oocytes or embryos, making polar body-based PGT-MT ineffective (Dean *et al.*, 2003; Sato *et al.*, 2005; Gigarel *et al.*, 2011; Vandewoestyne *et al.*, 2011, 2012; Neupane *et al.*, 2014b; Ji *et al.*, 2021).

Significantly more data are available from cleavage-stage biopsy, typically performed on eight-cell embryos (Steffann *et al.*, 2006; Thorburn *et al.*, 2009; Gigarel *et al.*, 2011; Monnot *et al.*, 2011; Sallevelt *et al.*, 2013, 2015, 2017b,c; Spath *et al.*, 2021). Outcomes of children born after PGT-MT at the cleavage stage are generally positive, with postnatal mutation measurements aligning with those observed in embryonic cells, irrespective of the mutation analysed. In cases where embryos that tested negative for the analysed mutation were transferred, the mutation was always undetectable in the child (Steffann *et al.*, 2006; Sallevelt *et al.*, 2013). Similarly, when a heteroplasmic embryo was transferred, the children's mutation loads closely resembled those found in the embryo (Thorburn *et al.*, 2009; Monnot *et al.*, 2011; Sallevelt *et al.*, 2017b).

In most cases, two blastomeres were sampled, deviating from the typical procedure in conventional PGT, where usually only one blastomere is taken (Carvalho *et al.*, 2020a,b; Kokkali *et al.*, 2020). While most studies show minimal variation in mutant loads between sister blastomeres (Dean *et al.*, 2003; Neupane *et al.*, 2014b), some embryos occasionally exhibit significant differences, with up to a 24% variation reported (Steffann *et al.*, 2006; Gigarel *et al.*, 2011; Vandewoestyne *et al.*, 2012; Sallevelt *et al.*, 2013, 2017b).

Removing two cells is highly invasive and has been demonstrated to markedly reduce embryo viability (Goossens *et al.*,

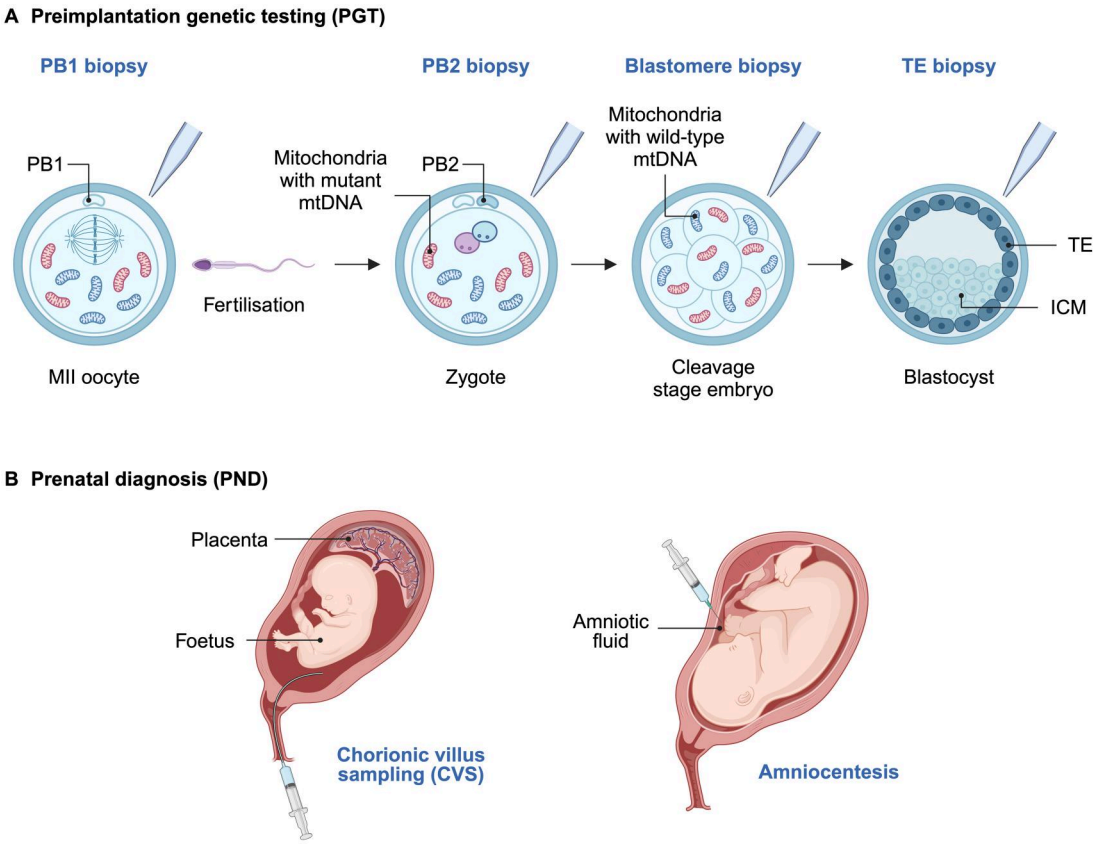


Figure 3. Reproductive genetic testing techniques. (**A**) Embryo biopsy for preimplantation genetic testing (PGT) can be performed at three stages of preimplantation development. First, at the metaphase II (MII) oocyte and/or the zygote stage by analysing the first and/or second polar body (PB1/2), respectively. Second, at the cleavage stage (normally when the embryo is eight cells) by analysing one or two blastomeres. Third, at the blastocyst stage by analysing trophectoderm (TE) cells. (**B**) In prenatal diagnosis (PND), chorionic villus sampling (CVS) involves using a fine suction tube, which is inserted through the cervix, to remove chorionic villi from the placental tissue. During amniocentesis, a needle is inserted transabdominally to withdraw foetal cells from the amniotic fluid. Following all procedures, (mt)DNA is extracted from collected cells, and genetic analysis is performed to assess the presence and level of mutations, which helps predict the risk of serious disease in the child. ICM, inner cell mass; mtDNA, mitochondrial DNA. Figure created with BioRender.

2008; De Vos *et al.*, 2021). Consequently, there is increasing interest in the use of single-cell biopsy, despite its potential for misdiagnosis due to outlier results (Poulton *et al.*, 2019). Data suggest that these risks may be relatively low for the m.3243A>G mutation, making single blastomere biopsy sufficient for accurate PGT-MT (Sallevelt *et al.*, 2017c).

A procedure known for its significantly lower invasiveness is TE biopsy, performed at the blastocyst stage. This process involves sampling approximately 5–10 cells from the outer TE layer, leaving the ICM intact. This method is considered safer for embryonic development compared to cleavage-stage biopsy (Scott *et al.*, 2013) and may enhance the accuracy of PGT-MT due to the simultaneous analysis of more cells (Treff *et al.*, 2012; Heindryckx *et al.*, 2014; Smeets *et al.*, 2015).

Though data on the representativeness of a TE biopsy are limited, available studies in humans do suggest that, in general, there is good agreement between individual TE biopsies taken from the same embryos (Treff *et al.*, 2012; Heindryckx *et al.*, 2014), as well as between the TE and the ICM (Treff *et al.*, 2012; Ji *et al.*, 2021). However, some discrepancies have been observed with up to 23% heteroplasmy difference reported in a single blastocyst (Heindryckx *et al.*, 2014). Conflicting results have also been reported in animal studies. While analysis of heteroplasmic mice indicated that TE sampling is largely representative of the whole blastocyst (Neupane *et al.*, 2014b), a greater disparity in heteroplasmy levels between TE cells and the ICM was observed in non-human primates (Lee *et al.*, 2012).

To date, the clinical application of TE analysis has been reported in only four studies (Treff *et al.*, 2012; Heindryckx *et al.*, 2014; Ji *et al.*, 2021; Spath *et al.*, 2021). In two of these, a single embryo that tested negative for the mutation was transferred, resulting in the birth of healthy, mutation-free children, as confirmed by post-delivery testing (Heindryckx *et al.*, 2014; Spath *et al.*, 2021). The other studies involved the transfer of heteroplasmic embryos rather than mutation-free ones (Treff *et al.*, 2012; Ji *et al.*, 2021). Ji *et al.* (2021) reported the transfer of a euploid blastocyst with a mutation load of 53.9%. A pregnancy was achieved but was terminated due to cleft lip and palate. Importantly, the mutation load in various tissues of the foetus was consistent with that of the TE biopsy and amniotic fluid. On the other hand, in the case study reported by Treff *et al.* (2012), a blastocyst with a 12% mutation load was transferred. Initial post-delivery tests described a slight increase in mutation load (15%), but later tests varied, showing either undetectable levels or below 10%. Despite these findings, the child developed severe health issues. Follow-up tests from another laboratory indicated higher mutation levels of 42–52% (Mitalipov *et al.*, 2014).

These findings raise questions about the reliability of PGT-MT in accurately reflecting the mutation load in the entire embryo. Technical issues could have also contributed to differences in

Table 1. Summary of published reports of children born after preimplantation genetic testing for mitochondrial DNA mutations (PGT-MT).

Mutation	Embryo stage and biopsy type	Number of embryos transferred	Result from biopsy	Result at birth	Analysed post-partum tissues	Notes	Reference
m.8993T>G	Cleavage—blastomere	2	0/0% and 0/0%	0%	Cord blood	Healthy at 7 months	Steffann *et al.* (2006)
m.8993T>G	Cleavage—blastomere	1	2.5%	4%	Cord blood	Healthy at 4.5 years	Thorburn *et al.* (2009)
m.8993T>G	Cleavage—blastomere	2	0/0% and 0/0%	0%	Cord blood	Healthy at birth	Sallevelt *et al.* (2013)
m.8993T>G	Blastocyst—TE	1	0%	0%	Cord blood and cheek swab		Spath *et al.* (2021)
m.8344A>G	Cleavage—blastomere	2	53% and 59%	63%	Not specified		Steffann *et al.* (2014)
m.14487T>C	Cleavage—blastomere	2	5.4/6.1% and 12.2/12.3%	5%	Cord blood	Healthy at 7 months	Sallevelt *et al.* (2017b)
m.3243A>G	Cleavage—blastomere	2	5/5% and 13/16%	5 ± 1% 15 ± 5% 4 ± 0%	Cord blood Placenta Amniotic fluid		Monnot *et al.* (2011)
m.3243A>G	Blastocyst—TE	1	12%	15%	Cheek swab (1 month)		Treff *et al.* (2012)
				0%	Cheek swab and urine (5 and 12 months)		
				<10%	Blood (5 and 12 months)		
				47% and 52%	Blood and urine (6 weeks)	Child showed neurological and developmental problems	Mitalipov *et al.* (2014)
				46% and 42%	Blood and urine (18 months)		
m.3243A>G	Blastocyst—TE	1	0%	0%	Blood, urine, and saliva		Heindryckx *et al.* (2014)
m.3697G>A	Blastocyst—TE	1	53.9%	47.7 ± 1.8% 49.7%	Induced foetus Amniotic fluid	Pregnancy terminated due to cleft lip and palate	Ji *et al.* (2021)

(…/…) indicates heteroplasmy levels in each of the two blastomeres analysed. TE, trophectoderm.

observed results, as noted in various reports (Mitalipov *et al.*, 2014; Smeets *et al.*, 2015; Poulton *et al.*, 2019). However, the method used in the PGT-MT case described by Treff *et al.* (2012) was rigorously validated and deemed reliable. This makes it unlikely that technical variations alone could account for significant discrepancies between laboratory results. Additionally, the child's phenotype in this case did not correspond to the typical presentation of the m.3243A>G mutation, suggesting that the clinical issues observed may not have been related to the mtDNA mutation.

The optimal choice between blastomere and TE biopsy remains uncertain due to limited data. Currently, most PGT-MT providers prefer blastomere biopsy, in line with the ESHRE good practice recommendations (Carvalho *et al.*, 2020a,b), but interest in TE-based approaches is growing. Over the last decade, TE biopsy has become the dominant methodology for all other forms of PGT due to its advantages in terms of the preservation of embryo viability and the ability to combine testing for specific inherited disorders with analyses of spontaneously occurring genetic abnormalities that impact the chances of a successful pregnancy (e.g. aneuploidy) (Spath *et al.*, 2021). Although the impact of TE biopsy on embryos with poor morphology is not well understood, it is generally considered less invasive than cleavage-stage biopsy.

Challenges and considerations in embryo selection: setting 'safe' mutation load thresholds

The goal of PGT-MT is to transfer embryos without detectable mutant mtDNA, but when this is not possible, embryos with mutations below the disease threshold may be considered (Poulton and Bredenoord, 2010; Poulton *et al.*, 2019). Determining a 'safe' mutation load threshold is complex and varies among providers. A general threshold of 18% has been proposed, based on pooled data from numerous mtDNA mutations across a large patient cohort (Hellebrekers *et al.*, 2012). For well-studied mutations, however, thresholds are best calculated individually. For instance, for the m.8993T>G mutation, 30% is typically used (Sallevelt *et al.*, 2013, 2015), due to a strong correlation between mutation load and disease severity, with well-documented predictive ranges (Mäkelä-Bengs *et al.*, 1995; White *et al.*, 1999a; Carelli *et al.*, 2002). This mutation shows a stable and uniform distribution across tissues during intrauterine development (Harding *et al.*, 1992; Ferlin *et al.*, 1997; Steffann *et al.*, 2007, 2021) and adulthood (White *et al.*, 1999b; Carelli *et al.*, 2002; Steffann *et al.*, 2007).

Conversely, for the m.3243A>G mutation, thresholds ranging from 15% to 30% are commonly applied because of its weak correlation with disease severity (Gigarel *et al.*, 2011; Sallevelt *et al.*, 2015). The mutation's instability over time and the uneven tissue distribution further complicate the prediction of 'safe' mutation thresholds (Chinnery *et al.*, 1999; Frederiksen *et al.*, 2006). For rare private mutations, predicting a threshold is often impossible due to insufficient data on the relationship between mutation load and phenotype (Poulton and Bredenoord, 2010).

In practice, patients may be given more autonomy in embryo selection (Monnot *et al.*, 2011; Treff *et al.*, 2012; Spath *et al.*, 2021), which may involve transferring embryos with higher mutation loads or opting to only transfer mutation-free embryos to minimize risks. Patients must be advised of the associated risks,

especially when all the embryos produced have mutation levels in a range that might potentially produce a phenotype; children should be closely followed up, and thresholds should be revised as new data emerge.

Additionally, in cases of heteroplasmic mtDNA mutations, transferring male embryos may also be considered, as they would not pass on the mutation to future generations.

Prenatal diagnosis (PND)

PND for mtDNA mutations (PND-MT) is carried out after pregnancy has occurred, usually in the first or second trimester, using one of two methods. Chorionic villus sampling (CVS), performed between the 10th and 14th week of pregnancy, involves the sampling of placental tissue, specifically the chorionic villi. Amniocentesis, typically carried out from the 15th week of pregnancy onwards, involves the collection of amniotic fluid, which contains foetal cells (Alfirevic *et al.*, 2017). The mtDNA in these samples is analysed to detect and quantify mutations, helping predict the risk of serious disease in the child. Based on this evaluation, the patient may decide whether to proceed or terminate the pregnancy (Harding *et al.*, 1992; Bredenoord *et al.*, 2008b) (Fig. 3B).

Like PGT-MT, PND-MT can only minimize the risk of disease transmission but cannot eliminate this possibility. The predictive power depends on whether the mutation load measured in the prenatal sample is representative of that in the foetus. The accuracy of this prediction relies on the stability of the foetal mutation load over time, therefore reflecting the mutation load in the newborn, and on whether the risk of postnatal disease can be accurately inferred from the measured mutation levels (White *et al.*, 1999b; Poulton and Turnbull, 2000; Smeets *et al.*, 2023).

Meeting these requirements is largely dependent on the specific mutation and available knowledge of it. Although global experience with PND-MT is greater than with PGT-MT, it remains limited. Hence, the predictive value of PND-MT remains uncertain, and it should be considered a risk-reduction strategy (Steffann *et al.*, 2021).

Optimal timing and method for PND-MT: factors to consider

There is limited understanding of the segregation of wild-type and mutant mtDNA during foetal somatic tissue development. However, aggregated data indicate that tissue homogeneity and temporal mutation load stability are largely preserved during *in utero* development, possibly independent of the mutation type. Studies on foetal tissues and organs, including those from pregnancies terminated or miscarried up to 25 weeks of gestation, show stable mutation levels across tissues. Importantly, this observation holds true for both the m.8993T>G (Harding *et al.*, 1992; Ferlin *et al.*, 1997; Steffann *et al.*, 2007, 2021) and the m.3243A>G mutations (Matthews *et al.*, 1994; Cardaioli *et al.*, 2000; Monnot *et al.*, 2011; Vachin *et al.*, 2018; Steffann *et al.*, 2021). Furthermore, analyses of amniotic fluid collected at different stages of pregnancy show stable mutation levels of m.8993T>G (Steffann *et al.*, 2007, 2021) and m.3243A>G (Bouchet *et al.*, 2006; Monnot *et al.*, 2011). Similar findings were also noted for rarer mutations such as m.14459G>A and m.14487T>C (Steffann *et al.*, 2021). Information on private and less-studied mutations is lacking.

The question of which method is best suited for PND-MT is complex. CVS, performed in the first trimester, allows for earlier pregnancy termination if the result is unfavourable. However, CVS analyses placental tissue, which might not fully represent the foetal mutation load. Conversely, amniocentesis, conducted in the second trimester, involves sampling amniotic fluid containing foetal epithelial cells, also known as amniocytes, potentially offering a more representative view of the foetal mutation load, although it has the significant drawback of later-stage pregnancy termination if needed (Smeets *et al.*, 2023). Given that heteroplasmic mtDNA mutation levels can change when cells are grown *in vitro*, it is important for prenatal samples, whether chorionic villi or amniocytes, to be tested directly and not cultured.

Studies have shown that mutation loads measured in amniotic fluid are comparable to those found in foetal tissues (Steffann *et al.*, 2007, 2021). Similarly, neonatal samples were consistent with those observed in amniotic fluids, regardless of the mutation tested (Chou *et al.*, 2004; Monnot *et al.*, 2011; Vachin *et al.*, 2018; Steffann *et al.*, 2021). Cumulative data also suggest that CVS offers a relatively accurate estimate of foetal or neonatal mutation load in most cases (Harding *et al.*, 1992; Matthews *et al.*, 1994; Ferlin *et al.*, 1997; Steffann *et al.*, 2007, 2021; Monnot *et al.*, 2011; Nesbitt *et al.*, 2014; Vachin *et al.*, 2018). Nonetheless, exceptions have been recorded. Two PND-MT cases demonstrated discrepancies between CVS and neonatal cord blood findings (Vachin *et al.*, 2018). In the first case (m.3243A>G), the CVS mutation load was 16% higher than the post-birth values, while in the second case (m.9185C>T), it was 24% higher. Interestingly, analysis of numerous individual placental biopsies revealed extensive intra-tissue variation, reaching up to ~35% (n = 12) in the first case and ~43% (n = 37) in the second case. However, the mean placental mutation loads corresponded to postnatal measurements in both children (Vachin *et al.*, 2018).

Non-uniform segregation of mtDNA in the placenta has been documented previously (Marchington *et al.*, 2006, 2010), likely influenced by the mutation type and possibly resembling patterns observed in adult tissues. For instance, m.8993T>G tends to display uniform tissue distribution postnatally, whereas m.3243A>G can exhibit significant variability across adult organs and tissues. A tendency towards mosaicism might influence the accuracy of CVS-based PND-MT for specific mutations.

Additional factors impacting the accuracy of PND-MT include the age of the foetus, the overall mutation load, and the size of the placental sample. While the placenta as a whole seems to accurately reflect the mutant load in postnatal tissues, smaller samples may not fully capture placental mosaicism, leading to potential discrepancies (Marchington *et al.*, 2006, 2010). This diagnostic challenge might be mitigated by the parallel evaluation of two distinct biopsies or undertaking CVS followed by amniocentesis at a later stage (Vachin *et al.*, 2018). Such strategies might be particularly valuable for mutations known to have the potential for non-uniform distribution, such as m.3243A>G (Steffann *et al.*, 2021; Smeets *et al.*, 2023).

Deciding when to terminate: interpreting results of PND-MT

Interpreting PND-MT test results involves understanding the correlation between heteroplasmy levels and disease manifestation. Results are typically classified into three categories: low risk (zero or low mutation load), high risk (high mutation load), and intermediate risk (uncertain due to insufficient data). Thresholds for categorizing risk levels should be informed by established genotype-phenotype correlations. PND-MT can provide meaningful information about the child's disease risk for mtDNA mutations with well-established correlations between mutation load and disease occurrence or severity, such as m.8993T>G. However, predicting this risk may be significantly more challenging for mtDNA mutations for which there are no clear

correlations (e.g. m.3243A>G) or for individual mtDNA mutations that have not been extensively studied (Mavraki *et al.*, 2023).

Clinical utility of reproductive genetic testing

Choosing between PGT-MT and PND-MT depends on mutation type, hereditary risk, and patient circumstances. For de novo mtDNA mutations or single large-scale deletions, recurrence risk is low, ~4% (Chinnery *et al.*, 2004; Sallevelt *et al.*, 2017a). In such instances, reproductive genetic testing may be carried out primarily for reassurance purposes. PND-MT would usually be the preferred choice when recurrence risk is low, as the probability of a mutation-free foetus is high, minimizing potential difficulties in interpreting test results and arguing against the institution of a complex clinical pathway (Bredenoord *et al.*, 2008b; Smeets *et al.*, 2015, 2023; Sallevelt *et al.*, 2017c; Mavraki *et al.*, 2023). Nevertheless, some may choose PGT-MT for reassurance or due to existing fertility treatments (Sallevelt *et al.*, 2017c; Smeets *et al.*, 2023).

Although asymptomatic women with low levels of pathogenic mutations have a low risk of disease transmission, reproductive genetic testing is nonetheless recommended. Both PGT-MT and PND-MT are viable options, with PGT-MT likely yielding transferable embryos while PND-MT remains an appropriate choice given the reduced risk of having an affected child (Mavraki *et al.*, 2023; Smeets *et al.*, 2023). In contrast, women carrying elevated levels of mtDNA mutations face significant reproductive challenges (Poulton *et al.*, 2019; Mavraki *et al.*, 2023). PGT-MT is preferred, though it may require several ART cycles to find embryos with 'safe' mutation loads (Pickett *et al.*, 2019; Steffann *et al.*, 2021; Mavraki *et al.*, 2023; Smeets *et al.*, 2023). PND-MT may be particularly difficult due to the increased likelihood of intermediate or high foetal mutation levels, which can be difficult to interpret. Additionally, there is a risk of having to manage multiple pregnancy terminations (Mavraki *et al.*, 2023; Smeets *et al.*, 2023).

Reproductive genetic testing is generally not suitable for women with homoplasmic mtDNA mutations, as all embryos would be affected. However, some homoplasmic mutations, such as those associated with LHON, show variable and sex-related disease expression. Male carriers face a roughly 50% risk of developing the disease, whereas, for women, the risk is significantly lower at ~10% (Black *et al.*, 1995; Yu-Wai-Man *et al.*, 2002). This equates to a 30% overall risk of an affected child for each pregnancy for female carriers (5% for daughters and 25% for sons). For these patients, considering PGT to determine chromosomal sex and prioritizing female embryos for transfer could be an option, reducing the recurrence risk from 30% to 10% (Bredenoord *et al.*, 2008a; Poulton and Bredenoord, 2010).

Germline nuclear transfer (NT)

NT, also known as mitochondrial replacement therapy (MRT) or mitochondrial donation, has emerged as a promising option for women carrying homoplasmic mtDNA mutations or producing only oocytes with high mutation loads (Greenfield *et al.*, 2017). This procedure involves transferring the nuclear genetic material in the form of a karyoplast (chromosomes and/or nucleus surrounded by a small amount of cytoplasm enclosed in plasma membrane) from a prospective mother's oocyte or zygote to an enucleated counterpart (called cytoplast) from a healthy donor (Fig. 4). This innovative IVF treatment, often referred to as 'three-parent baby' in the media, offers women with pathogenic mtDNA mutations the chance to have genetically related children whilst reducing the risk of mtDNA disease (Amato *et al.*, 2014). It is estimated that ~150 women in the UK and 780 in the USA could benefit from NT each year (Gorman *et al.*, 2015a).

In 2015, the UK became the first country to approve the clinical use of NT to prevent mtDNA disease transmission (Craven *et al.*, 2016), followed by Australia in 2022 (Allen *et al.*, 2024). Clinical pathways have been established for women with mtDNA disease to ensure the best possible care for patients (Gorman *et al.*, 2018; Poulton *et al.*, 2019). However, the legality of NT varies worldwide: for instance, the USA restricts it (Cohen *et al.*, 2020), while Greece and Ukraine have conducted successful NT clinical trials for infertility treatment (Mazur, 2020; Costa-Borges *et al.*, 2023). Currently, ESHRE's good practice recommendations advise the use of NT only for research purposes (Lundin *et al.*, 2023). The legal aspects of NT will be further discussed in the *Legal and ethical considerations* section.

The different stages of oocyte maturation offer several opportunities for the application of NT, which can be performed either before or after fertilization (Wolf *et al.*, 2015). In unfertilized oocytes, maternal spindle transfer (MST), first polar body transfer (PB1T), or germinal vesicle transfer (GVT) can be performed. Alternatively, the nuclear genome can be transferred from zygotes using pronuclear transfer (PNT) or second polar body transfer (PB2T) (Craven *et al.*, 2017) (Fig. 4). The most important outcomes when evaluating the efficacy of the technique are embryo development, genetic normality, and minimal maternal mtDNA transfer within the karyoplast, as well as normal health and growth in offspring born (Greenfield *et al.*, 2017).

Following transfer of the nuclear genome, fusion with the donor's cytoplasm can be achieved by various means, with the most common ones being the viral envelope from the haemagglutinating virus of Japan (HVJ-E) (also known as inactivated Sendai virus) and electrofusion (Tachibana *et al.*, 2009, 2010). Methods such as microinjection and polyethylene glycol (PEG) have also been described but have been less frequently used (Tesarik *et al.*, 2000).

Efficacy of NT: methodologies and supporting evidence

Maternal spindle transfer (MST)

MST involves transferring the MII spindle-chromosome complex (MII-SCC) from a mature MII oocyte to the enucleated cytoplast of a healthy donor oocyte (Tachibana *et al.*, 2013) (Fig. 4). This process requires polarized light birefringence microscopy for visualizing the meiotic spindle (Oldenbourg *et al.*, 1998; Liu *et al.*, 2000b) and the use of cytoskeletal inhibitors, such as cytochalasin B or D, to manipulate the cytoplasm while minimizing damage (Tachibana *et al.*, 2010). After reconstruction, the oocyte is fertilized with sperm via ICSI to initiate embryo development.

The potential of MST has been demonstrated in various animal models, including mouse studies, which revealed high fertilization rates, live births, and minimal mtDNA heteroplasmy in the offspring (Wang *et al.*, 2001; Neupane *et al.*, 2014a; Tang *et al.*, 2019; Costa-Borges *et al.*, 2020). Research in rhesus macaques *(Macaca mulatta)* indicated successful fertilization, viable embryos, and the birth of healthy monkeys following MST (Tachibana *et al.*, 2009, 2013; Lee *et al.*, 2012). Follow-up reports showed that offspring remained healthy, without mitochondrial dysfunction, and were fertile (Tachibana *et al.*, 2013; Ma *et al.*, 2021).

Subsequent investigations in human oocytes confirmed the feasibility of the technique (Paull *et al.*, 2013; Tachibana *et al.*, 2013). Although some abnormalities in fertilization were observed in a portion of MST zygotes, normally fertilized MST zygotes exhibited similar rates of blastocyst development and embryonic stem cell (ESC) derivation compared to controls (Tachibana *et al.*, 2013). MST has also been applied to oocytes from patients carrying pathogenic mtDNA mutations. Results

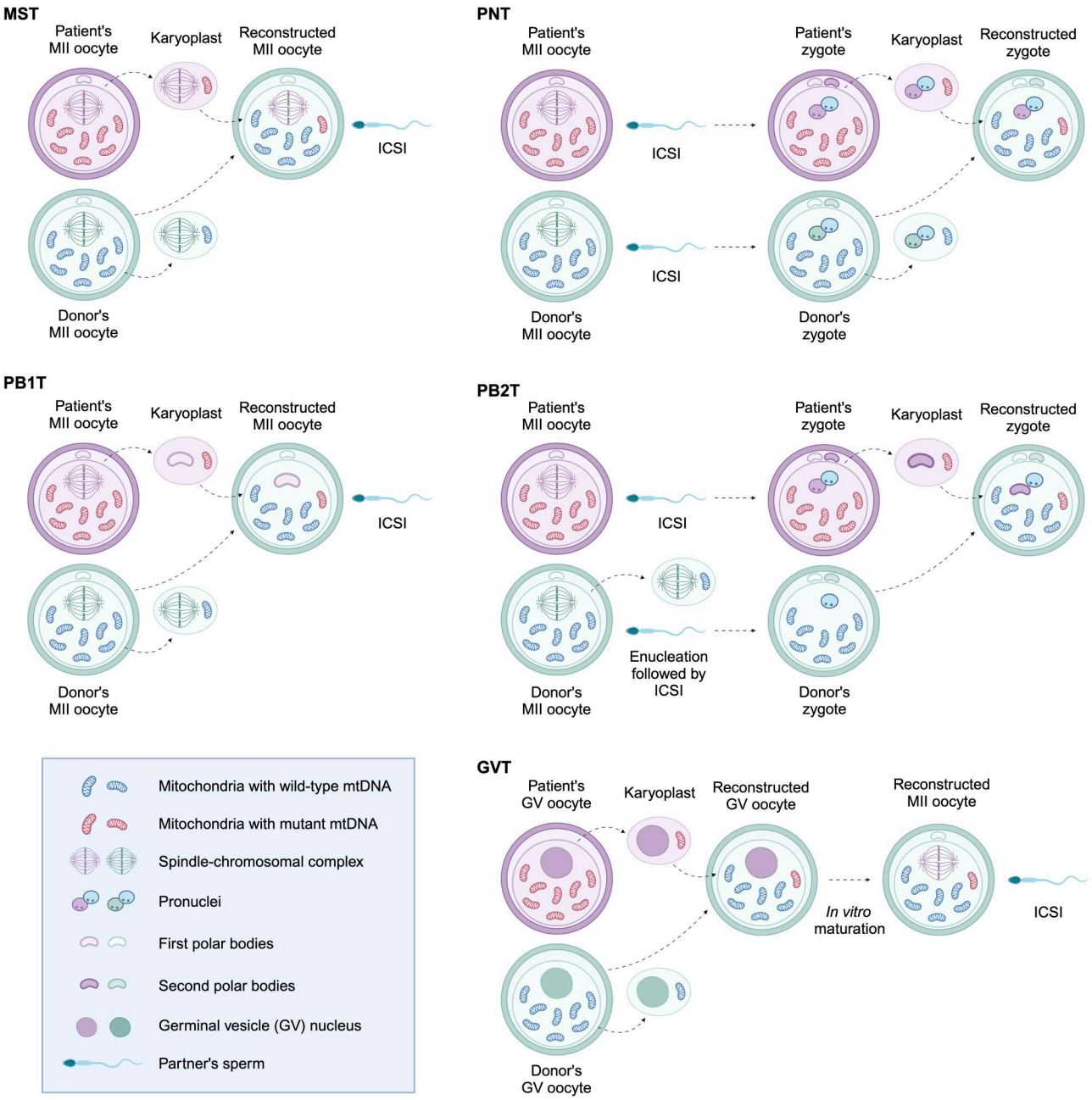


Figure 4. Schematic diagram of different nuclear transfer (NT) techniques. The procedures of maternal spindle transfer (MST), pronuclear transfer (PNT), first and second polar body transfer (PB1T and PB2T), and germinal vesicle transfer (GVT) are shown. The karyoplast is isolated from the patient's oocyte or zygote and then transferred into the recipient cytoplast from a healthy donor. Figure created with BioRender.

showed a compromised fertility potential in patients, though it was limited by a small sample size and high fertilization and blastocyst rates in reconstructed oocytes (Kang *et al.*, 2016). The potential of MST received additional validation through subsequent preclinical human studies (Yamada *et al.*, 2016; Tang *et al.*, 2022), also involving oocytes from transgender men (Christodoulaki *et al.*, 2023).

Notably, a ground-breaking study documented the first clinical application of MST to prevent transmission of mtDNA disease (Zhang *et al.*, 2017a), resulting in the birth of a baby boy with low (subclinical) levels of mtDNA mutation in each of the tissues analysed. Unfortunately, later follow-up data concerning heteroplasmy levels and the health of the child are not currently available. Recent research has examined whether the use of MST to replace the whole cytoplasm of an oocyte could address infertility in women with poor oocyte quality and a history of failed IVF attempts. The pilot study described encouraging results, with six healthy births reported (Costa-Borges *et al.*, 2023) (Table 2).

Pronuclear transfer (PNT)

PNT consists of transferring both maternal and paternal pronuclei between zygotes prior to syngamy (Fig. 4). Similarly to MST, micromanipulation of the zygote requires the use of cytoskeletal inhibitors to minimize any potential damage to the zygote. PNT was developed four decades ago by McGrath and Solter (1983) and laid the groundwork for various NT techniques. This pioneering research demonstrated the success of PNT by producing healthy murine offspring that could reproduce normally (McGrath and

Table 2. Summary of the included studies on maternal spindle transfer (MST) in animal and human models.

Species	Materials	Fusion method and efficiency	Developmental potential	Live birth rate	mtDNA carryover	Notes	Study
Mouse	Oocytes from Kunming and C57BL/6J mice	EF: 17.6% (25/142)	Not indicated	25% (2/8)	Not tested	Body weight of MST offspring similar to controls; fertilization with IVF	Wang *et al.* (2001)
	Oocytes from NZB/OlaHsd* and B6D2/F1 mice	HVJ-E: 85.2% (23/27)	Blastocyst rate: 82.6% (19/23) in parthenotes, similar to controls (85.2%), and 0% (0/12) for ICSI	Not tested	Embryos: 0.29 ± 0.63%, undetectable in 17/24	Parthenogenic activation and ICSI	Neupane *et al.* (2014a)
	Oocytes from NZB/OlaHsd* and B6D2F1 mice	HVJ-E: 90.2% (74/82)	Blastocyst rate: 81.8% (36/44), similar to controls (86.9%)	Not tested	Embryos: 3.57% ± 3.39%		Tang *et al.* (2019)
	Oocytes from NZB/OlaHsd* and B6CBA/F1 mice	HVJ-E: 95.7% (228/238)	Blastocyst rate: 51.4% (109/212), similar to B6CBA/F1 controls (77.8%) and higher than NZB/OlaHsd controls (5.6%)	21.5% (14/65), similar to controls (25.9%)	PB from two-cell embryos: 6.2 ± 6.2%, higher than embryo; blastomere of eight-cell embryos: 1.3 ± 1.0%, similar to embryo; TE biopsy: 1.7 ± 0.9%, similar to embryo; offspring: 2.3 ± 1.3%		Costa-Borges *et al.* (2020)
Non-human primate	Oocytes from non-human primates (*Macaca mulatta*)	HVJ-E and EF: 90% (78/87) and 73% (11/15), respectively	Blastocyst rate: 61% (45/74) for HVJ-E, similar to controls (60%) and 9% (1/11) for EF, lower than controls; ESC derivation: 25%, similar to control	26.7% (4/15), similar to controls	Infants: 3%; ESCs: undetectable Adults: ≤3% in three monkeys; heteroplasmy up to 17% in one monkey; paternal mtDNA up to 33% in two monkeys	First live births in non-human primates MST rhesus macaques reached healthy adulthood and were fertile	Tachibana *et al.* (2009) Ma *et al.* (2021) (follow-up study)
Human	Donated MII and *in vitro* matured oocytes	PEG: 87% (40/46) or microinjection 83% (29/35)	Not indicated	Not tested	Not tested	Parthenogenic activation	Tesarik *et al.* (2000)
	Donated MII oocytes	HVJ-E: fusion rate not indicated	Blastocyst rate: 62% (13/21) for 2PN zygotes, similar to controls (76%); 6 ESC lines derived (32%)	Not tested	Embryos: 0.5 ± 0.4%; ESCs: 0.6 ± 0.9%	Abnormal fertilization in 52% of MST embryos	Tachibana *et al.* (2013)

(continued)

Table 2. (continued)

Species	Materials	Fusion method and efficiency	Developmental potential	Live birth rate	mtDNA carryover	Notes	Study
	Donated MII oocytes	HVJ-E and EF: fusion rate not indicated	Blastocyst rate: 38.8% (7/18) compared to 52% (54/103) in control IVF group and 33% (7/21) in control parthenogenic group; 4 ESC lines derived (15%)	Not tested	Embryos: 0.31 ± 0.27%; ESCs: 0% (P2–59) in three lines; 2.79 ± 0.27% (P4–10), then 0% (from P14) in one line	Prevention of oocyte activation by spindle cooling; parthenogenic activation	Paull *et al.* (2013)
	Donated MII oocytes from healthy donors and from carriers of mtDNA variants (m.8993T>G; m.13513G>A; m.3243A>G)	HVJ-E: fusion rate not indicated	Blastocyst rate: 62.5% (20/32) in healthy oocytes and 50% (6/12) in oocytes from mutation carriers; 18 ESC lines derived	Not tested	Embryos: <1%; ESCs: ≤1% in 13 control MST lines, while one control MST line increased from 81% to 100%; ≤1% in two patient MST lines, while one patient MST line increased from 4% to 8%		Kang *et al.* (2016)
	Donated MII oocytes	HVJ-E: fusion rate not indicated	Blastocyst rate: 32% (fresh cytoplast and cryo-karyoplast) compared to 22% in controls with fresh oocyte and 9% in controls with cryo-oocytes; 8 ESC lines derived (25%)	Not tested	Embryos: 0.2%; ESCs: 0.42% in seven lines, while one line increased from 1.3% (P1) to 53.2% (P36)	Parthenogenic activation	Yamada *et al.* (2016)
	MII oocytes from a m.8993T>G carrier	EF: 100% (5/5)	Blastocyst rate: 100% (4/4); one male euploid embryo	100% (1/1)	Blastocyst: 5.10 ± 1.11%; tissue from new-born: 1.60 ± 0.92%	First human live birth following MST	Zhang *et al.* (2017a)
	Donated MII oocytes from a m.11778G>A carrier	HVJ-E: 100% (5/5)	Cleavage rate: 100% (5/5); blastocyst rate: 20% (1/5)	Not tested	Blastocysts: 2.9% ± 1.1%		Tang *et al.* (2022)
	In vitro matured donated oocytes from transgender men	HVJ-E: fusion rate not indicated	Blastocyst rate: 20.6% (6/29), similar to controls (25.4%)	Not tested	Not tested		Christodoulaki *et al.* (2023)
	MII oocytes from patients with idiopathic infertility	HVJ-E: 91.8% (112/122)	Blastocyst rate: 62.4% (53/85); 45.2% (24/53) were euploid	31.6% (6/19)	Embryos: ≤1%; infants: ≤1% in five children, while one kid showed 40–60% maternal haplotype		Costa-Borges *et al.* (2023)

MST, maternal spindle transfer; EF, electrofusion; HVJ-E, envelope of haemagglutinating virus of Japan; PEG, polyethylene glycol; MII, metaphase II; PB, polar body; ESC, embryonic stem cells; PNs, pronuclei.
* Model for embryo developmental arrest.
Blastocyst rate over fertilization rate. If not indicated, fertilization is performed with ICSI.

Solter, 1983). Subsequent investigations extended the applicability of PNT to study mtDNA segregation in heteroplasmic mice (Meirelles and Smith, 1997) and demonstrated its potential in preventing transmission of mtDNA disease in a mouse model with a large-scale mtDNA deletion (ΔmtDNA) (Sato *et al.*, 2005). Recently, PNT was successfully used to rescue poor embryo development of *in vitro* matured mouse oocytes (Christodoulaki *et al.*, 2024).

The first attempt at PNT with human zygotes involved exchanging pronuclei between abnormally fertilized zygotes (Craven *et al.*, 2010). Further optimization focused on determining the compatibility of PNT with development to the blastocyst stage, using normally fertilized (2PN) zygotes. The modified protocol, comprising early PNT 8-h post-fertilization, significantly improved zygote survival and resulted in a notably enhanced blastocyst formation similar to non-manipulated zygotes (Hyslop *et al.*, 2016). Another study aimed to assess the efficiency of PNT in preventing transmission of the pathogenic m.11778G>A mtDNA mutation (Tang *et al.*, 2022).

In 2023, the Human Fertilisation and Embryology Authority (HFEA), the UK fertility regulator, confirmed in the news the first live births (fewer than five) following the application of PNT (Callaway, 2023). However, no further information about the procedure or the children has been disclosed, and it is not known whether the intervention has proven successful. Currently, this remains the only reported clinical use of PNT aimed at preventing transmission of mtDNA diseases. However, other clinics have applied the technique to address infertility. For instance, PNT has been used clinically to overcome embryo developmental arrest in one patient (Zhang *et al.*, 2016), and a fertility clinic in Ukraine reported 10 live births after using PNT to address repeated IVF cycle failures (Mazur, 2020). These results, presented in an online webinar, await publication in a peer-reviewed journal (Table 3).

Polar body transfer (PBT)

There are two variations of PBT, namely PB1T and PB2T, which can be viewed as complementary approaches to MST and PNT, respectively. In PB1T, the first polar body is removed from a mature MII oocyte and then transferred to an oocyte with the meiotic spindle removed, followed by fertilization. Conversely, PB2T entails transferring the second polar body into a zygote containing solely the male pronucleus (Fig. 4). The paternal pronucleus can be obtained either by fertilizing the donor's oocyte with sperm and subsequently removing the maternal pronucleus (Wakayama *et al.*, 1997) or by enucleating a mature oocyte and then fertilizing it with the partner's sperm, resulting in the formation of a single pronucleus (Tang *et al.*, 2019).

PBT presents promising advantages over MST and PNT, potentially offering minimal transfer of mutant mtDNA, the absence of cytoskeletal inhibitors, and less invasive manipulations. However, it has been less rigorously studied, and there are some concerns over the integrity of nDNA given that polar bodies are not viable cells and begin to degrade after formation. Evidence from mouse studies demonstrated the ability of PBT to support normal development and produce healthy offspring (Evsikov and Evsikov, 1994; Wakayama *et al.*, 1997; Wakayama and Yanagimachi, 1998; Wang *et al.*, 2014) and permit the generation of ESC lines (Wakayama *et al.*, 2007; Wang *et al.*, 2011). Moreover, investigations in pigs (Wang *et al.*, 2011) and non-human primates (Wang *et al.*, 2021) further support the feasibility of PBT in diverse species.

Limited human data suggest suboptimal embryo development following PB1T (Ma *et al.*, 2017). Nevertheless, Zhang *et al.* (2017b) reported the successful application of PB1T to overcome embryo fragmentation and arrest in a woman with failed IVF cycles. PB2T was also tested, but all zygotes arrested at pronuclear formation, and no blastocysts were achieved (Zhang *et al.*, 2017b). Subsequently, successful human blastocyst development was reported from both PB1T and PB2T, with low aneuploidy rates and very low levels of mutated mtDNA (Wu *et al.*, 2017) (Table 4).

Germinal vesicle transfer (GVT)

GVT involves transferring the GV from an immature oocyte to an enucleated oocyte of the same maturation status, followed by *in vitro* maturation and fertilization with sperm (Darbandi *et al.*, 2017) (Fig. 4). Initially, GVT aimed at understanding oocyte meiosis and maturation (Liu *et al.*, 1999, 2003; Takeuchi *et al.*, 1999; Zhang *et al.*, 1999; Cui *et al.*, 2005). Studies showed that reconstructed embryos could develop to the blastocyst stage and result in viable offspring in mice (Liu *et al.*, 2000a; Cui *et al.*, 2005; Takeuchi *et al.*, 2005). It has also been explored as a strategy to prevent transmission of mtDNA diseases, with minimal transfer of mutated mtDNA observed in reconstructed mouse oocytes (Neupane *et al.*, 2014a). Reports of human GVT, albeit limited, indicate that GVT does not induce chromosomal errors in reconstructed oocytes, which are capable of maturation, fertilization, and early embryonic cleavage (Takeuchi *et al.*, 1999; Zhang *et al.*, 1999) (Table 5). GVT shows promise for women producing only immature oocytes, potentially offering a fertility treatment option. However, further research is required before clinical application due to the inefficiency of *in vitro* maturation of human oocytes (De Vos *et al.*, 2021).

Safety of NT: challenges and solutions

mtDNA carryover and the reversion phenomenon

The efficacy of NT procedures for reducing the risks of mtDNA transmission not only relies on successful embryo development but also on minimizing the amount of pathogenic mtDNA in the reconstructed embryo, and consequently, in the resulting child. This consideration stems from the nature of the NT procedure, which inevitably involves transferring a small amount of cytoplasm with the nuclear genome in the karyoplast. This cytoplasm contains mitochondria from the affected oocyte/zygote and has the potential to induce heteroplasmy by transferring mutated mtDNA, also known as mtDNA carryover, to the reconstituted embryo (Craven *et al.*, 2017). Studies have shown consistently low levels of mtDNA heteroplasmy (<2%) in most reconstructed embryos (Tachibana *et al.*, 2013; Kang *et al.*, 2016; Tang *et al.*, 2019; Costa-Borges *et al.*, 2023). These levels of heteroplasmy would not be expected to cause symptoms of mitochondrial disease as they fall below the threshold for disease expression. However, the potential expansion of mutant mtDNA during subsequent development remains a concern.

To investigate mtDNA carryover *in vitro*, ESC lines derived from human NT blastocysts were monitored for heteroplasmy shifts. Surprisingly, 20% of ESCs showed rapid reversion to homoplasmic levels of maternal mtDNA independent of donor mtDNA haplotype or sequence, even when initial carryover was below 5% (Hyslop *et al.*, 2016; Kang *et al.*, 2016; Yamada *et al.*, 2016). Similar findings were observed in NT studies with non-human primates (Lee *et al.*, 2012; Ma *et al.*, 2021) and mice (Meirelles and Smith, 1997; Neupane *et al.*, 2015), indicating genetic drift and unexpected dominance of one mtDNA haplotype. Additionally, the

Table 3. Summary of the included studies on pronuclear transfer (PNT) in animal and human models.

Species	Materials	Fusion method and efficiency	Developmental potential	Live birth rate	mtDNA carryover	Notes	Study
Mouse	Zygotes from C3H/Hej and C57BL/6J mice	HVJ-E: 99% (67/68)	Blastocyst rate: 96% (64/67), similar to controls (100%)	15.6% (10/64), similar to controls (14.7%)	Not tested	PNT offspring survived to adulthood and were fertile	McGrath and Solter (1983)
	Zygotes from C57BL/6 and NZB/BinJ mice	EF: fusion rate not indicated	Not indicated	Six pups	Reconstructed zygotes: 19% (16–23%); adult mice: 0–69%, one female with high levels of heteroplasmy		Meirelles and Smith (1997)
	Zygotes from '*mito*' (ΔmtDNA, 4696 bp deletion) and C57BL/6J mice	EF: fusion rate not indicated	Not indicated	Live birth rate: 28% (11/39), similar to controls (26.5%)	New-borns: 11% (6–21%); adults: 23% (5–44%)	Offspring free of disease and lived longer than '*mito*' mice	Sato *et al.* (2005)
	Zygotes from NZB/OlaHsd* and B6D2/F1 mice	HVJ-E: 88.9% (16/18)	Blastocyst rate: 87.5% (14/16), similar to controls	Not tested	Embryos: 0.29 ± 0.75%		Neupane *et al.* (2014a)
	Zygotes from NZB/OlaHsd* and B6D2F1 mice	HVJ-E: 92.1% (82/89)	Blastocyst rate: 79.3% (65/82), similar to controls (82.5%)	Not tested	Blastocysts: 4.13% ± 3.42%		Tang *et al.* (2019)
	In vivo and *in vitro* grown B6D2 oocytes	HVJ-E: fusion rate not indicated	Blastocyst rate: 86.9%, similar to controls (96.4%)	10.5%, similar to controls (10%)	Not tested	Fertilization with IVF	Christodoulaki *et al.* (2024)
Human	Abnormally fertilized donated zygotes (1PN/3PN)	HVJ-E: fusion rate not indicated	Eight-cell stage: 22.2% (8/36); blastocyst rate: 8.3% (3/36), lower than controls (17.6% for 1PN/3PN embryos and 32% for 2PN embryos)	Not tested	Embryos: 8.1% ± 7.6%; reduced to 1.68 ± 1.81% in following optimization experiments		Craven *et al.* (2010)
	Donated MII oocytes; fertilized by ICSI	HVJ-E: fusion rate not indicated	Zygote survival: 92% (36/39) in ePNT and 59% (23/39) in LtPNT; blastocyst rate: 61.9% (13/21) in autologous PNT, comparable to controls (63.3%), 40% (32/80) in heterologous PNT, lower than controls; 5 ESC lines derived	Not tested	Embryos: <5%; ESCs: <2% in four lines, while one line increased from 1.3% (P1) to 53.2% (P36)	ePNT optimization	Hyslop *et al.* (2016)
	Fertilized oocytes from a patient with two failed IVF cycles	EF: 87.5% (7/8)	Four-cell stage: 71.4% (5/7)	Implantation: 60% (3/5); births: 0% deaths by umbilical cord collapse	Foetal cells: undetectable (undetermined sensitivity)		Zhang *et al.* (2016)
	In vitro matured donated oocyte from a m.11778G>A carrier	HVJ-E: 100% (1/1)	Embryo reached the morula stage	Not tested	Embryo: 3.1%	MII oocytes used for MST	Tang *et al.* (2022)

PNT, pronuclear transfer; EF, electrofusion; HVJ-E, envelope of haemagglutinating virus of Japan; MII, metaphase II; ESC, embryonic stem cells; PNs, pronuclei; ePNT, early PNT; ltPNT, late PNT.
* Model for embryo developmental arrest.
Blastocyst rate over fertilization rate. If not indicated, fertilization is performed with ICSI.

Table 4. Summary of the included studies on polar body transfer (PBT) in animal and human models.

Species	Materials	Technique	Fusion method and efficiency	Developmental potential	Live birth rate	mtDNA carryover	Notes	Study
Mouse	Zygotes from B6D2F1 and C86F1 mice	PB2T	EF: 67%	Morula/blastocyst rate: 32% (26/81), lower than controls (95%)	Not tested	Not tested	*In vivo* zygote collection	Evsikov and Evsikov (1994)
	Zygotes from B6CBAF1 and DBA/2 mice	PB2T	EF: fusion rate not indicated	Blastocyst rate: 59.4% (19/32) for early, synchronous PB1T	56.7% (17/30)	Not tested	76.5% of PBT offspring survived to adulthood and were fertile; *in vivo* zygote collection	Wakayama *et al.* (1997)
	MII oocytes from B6D2F1, CH3, and CD1 mice	PB1T	Microinjection: 53.8% (92/171)	Two-cell stage: 93.7% (74/79)	50% (37/74)	Not tested	PBT offspring survived to adulthood and were fertile	Wakayama and Yanagimachi (1998)
	MII oocytes and zygotes from B6D2F1/2 and 129/Sv mice	PB1T	EF: fusion rate not indicated	Blastocyst rate: 21% (15/71) in B6D2F1/2 and 63% (22/35) in 129/Sv mice	Not tested	Not tested	Parthenogenic activation	Wakayama *et al.* (2007)
		PB2T		Blastocyst rate: 38% (12/32); 18 ESC lines established (33–40% derivation rate)			*In vivo* zygote collection	
	MII oocytes and zygotes from NZW and BDF1 mice	PB1T	HVJ-E: 76% (19/25)	Blastocyst rate: 82.3% (14/17), similar to MST controls (85.7%) and IVF controls (76.2%)	PB1T: 42.8% (6/14)	Offspring: undetectable in all pups	Comparison with MST and PNT	Wang *et al.* (2014)
		PB2T	HVJ-E: 93.3% (28/30)	Blastocyst rate: 53.6% (15/28), lower than PNT controls (81.3%)	40% (6/15), similar to controls (44.4–56.3%)	Offspring: 1.7% ± 2.8%; both lower than MST and PNT	Comparison with MST and PNT; *in vivo* zygote collection	
	MII oocytes from NZB/OlaHsd* and B6D2F1 mice	PB1T	HVJ-E: 95.4% (83/87)	Blastocyst rate: 66.6% (39/59)	Not tested	Embryos: 0.62% ± 0.56%, lower than other NT techniques	Comparison with MST and PNT; for PB2T, oocytes were enucleated and then fertilized (only paternal PN forms)	Tang *et al.* (2019)
		PB2T	HVJ-E: 91.2% (52/57)	Blastocyst rate: 48.4% (30/62), similar to controls (65.5–75.7%)		Embryos: 0.91% ± 0.94%, lower than other NT techniques		
Porcine	*In vitro* matured porcine oocytes	PB1T	Microinjection: 97.7% (125/128)	Not tested	Not tested	Embryos: 0.31 ± 0.27%; ESCs: 0% (P2–59) in three lines, 2.79 ± 0.27% (P4–10), 0% (from P14) in one line; iPSCs: 0.01 ± 0.04%	Frozen–thawed PBT	Wang *et al.* (2011)

(continued)

Table 4. (continued)

Species	Materials	Technique	Fusion method and efficiency	Developmental potential	Live birth rate	mtDNA carryover	Notes	Study
Non-human primate	MII oocytes from non-human primates (*Macaca fascicularis*)	PB1T	HVJ-E: fusion rate not indicated	Not indicated; 20 two- to eight-cell stage PB1T embryos transferred	10% (2/20), similar to literature controls	Infants: up to 8%		Wang *et al.* (2021)
Human	Donated MII oocytes	PB1T	HVJ-E: fusion rate not indicated	Blastocyst rate: 42% (8/19), lower than controls (75%); one ESC line derived (derivation rate lower than controls)	Not tested	Not tested	Genetic profiles of PBT embryos and ESCs comparable to controls	Ma *et al.* (2017)
	In vitro matured donated oocytes	PB1T	HVJ-E: fresh 91.7% (22/24) and frozen 88.9% (8/9)	Blastocyst rate: fresh 30.1% (4/13) and frozen 50% (3/6), similar to controls (43.8%)	Not tested	Not tested	Frozen–thawed PBT	Zhang *et al.* (2017b)
		PB2T	HVJ-E: 100% (17/17)	Blastocyst rate: 0% (0/17), lower than controls (43.8%)				
	In vitro matured donated oocytes	PB1T	HVJ-E: fusion rate not indicated	Blastocyst rate: PB1T: 28.4% (19/67), similar to controls (27.8%); five ESC lines derived	Not tested	PB1T: embryos: 0.26% (0.07–0.5%); ESCs: 0.15% (P5–P20)		Wu *et al.* (2017)
		PB2T		Blastocyst rate: 28.6% (14/49), similar to controls (28.9%); seven ESC lines derived		Embryos: 0.37% (0.06–0.7%); ESCs: 0.22% (P5–P20)		

PBT, polar body transfer; PB1T, first polar body transfer; PB2T, second polar body transfer; MST, maternal spindle transfer; PNT, pronuclear transfer; EF, electrofusion; HVJ-E, envelope of haemagglutinating virus of Japan; ESC, embryonic stem cells; PNs, pronuclei.

* Model for embryo developmental arrest.

Blastocyst rate over fertilization rate. If not indicated, fertilization is performed with ICSI.

Table 5. Summary of the included studies on germinal vesicle transfer (GVT) in animal and human models.

Species	Materials	Fusion method and efficiency	Maturation rate	Developmental potential	Live birth rate	mtDNA carryover	Notes	Study
Mouse	GV oocytes from CB6F1 mice	EF: 77.8% (144/185)	66% (95/144)	Not tested	Not tested	Not tested	All oocytes showed normal karyotype	Liu *et al.* (1999)
	GV oocytes from C57BL/6 and FVB/N mice	EF: 91.9% (376/409)	55–83% in different media	Activation rate: 30–90% in different media	Not tested	Not tested	Parthenogenic activation	Liu *et al.* (2000a)
	GV oocytes and zygotes from CB6F1 mice	EF: 91% on average across all groups	64.4% (311/483)	Two-cell stage: 90% on average across all groups	10% on average across all groups	Not tested	Sequencial GVT + MST/PNT. *In vivo* zygotes + parthenogenic activation	Liu et al. (2003)
	GV oocytes from Kunming mice	EF: 89.8–94.4% for study groups	93.5–97.9% for study groups, similar controls (98.0–99.1%)	Cleavage rate: 43.8–54.3% in study groups, similar to controls (47.8–51.8%)	Not tested	Not tested	83.1% (54/65) of oocytes from young mice showed normal karyotype, compared to 50.7% (33/65) of oocytes from old mice	Cui *et al.* (2005)
	GV oocytes from B6D2/F1 mice	EF: 67.2% (299/445)	76.2% (228/299)	Blastocyst rate: 21.2% (11/52), similar to controls (35%)	12.9% (17/132), similar to controls (19.3%)	Not tested	95.2% (20/21) of oocytes showed normal karyotype	Takeuchi *et al.* (2005)
	GV oocytes from B6D2/F1 and NZB/OlaHsd mice*	HVJ-E: 90% (35/39)	71% (25/35), similar to controls (74%)	Two-cell stage: 93% (13/14); blastocyst rate: 0% (0/14)	Not tested	Undetectable in all embryos analysed	Parthenogenic activation	Neupane *et al.* (2014a)
Human	Donated GV oocytes	EF: 72.8% (115/158)	61.7% (71/115)	Cleavage rate: 93.8% (15/16); one embryo arrested at eight-cell stage and one at morula stage.	Not tested	Not tested	78.6% (11/14) of oocytes showed normal karyotype	Takeuchi *et al.* (1999)
	Donated GV oocytes	EF: 63% (12/19)	58% (7/12), similar to controls (69%)	Not tested	Not tested	Not tested	80% (4/5) of oocytes showed normal karyotype	Zhang *et al.* (1999)

GVT, germinal vesicle transfer; MST, maternal spindle transfer; PNT, pronuclear transfer; EF, electrofusion; HVJ-E, envelope of haemagglutinating virus of Japan; GV, germinal vesicle.
* Model for embryo developmental arrest.
Blastocyst rate over fertilization rate. If not indicated, fertilization is performed with ICSI.

paternal mtDNA contribution was detected in certain tissues of NT offspring in monkeys (Ma et al., 2021).

Notably, mtDNA reversion (also referred to as 'reversal') has been observed in two babies conceived with NT to treat idiopathic infertility in two independent clinical studies conducted in Greece (Costa-Borges et al., 2023) and Ukraine (Mazur, 2020). Although the patients were not known to carry mtDNA mutations, the relative levels of mtDNA from the oocyte donor and the patient were assessed in the resulting embryos. In one case from each study, maternal mtDNA increased from <1% in the embryo to around 40–60% in one baby and 74% in the other (Mazur, 2020; Costa-Borges et al., 2023). As discussed earlier, instances of genetic drift have also been reported following PGT, leading to mitochondrial disease symptoms post-birth (Treff et al., 2012; Mitalipov et al., 2014).

The mechanisms underlying mtDNA genetic drift remain poorly understood. Some hypotheses have suggested that replicative advantage based on mtDNA sequence or haplogroup may play a role. Animal studies showed the preferential proliferation of one mtDNA sequence over another due to genetic divergence (Burgstaller et al., 2014). Positive selection of pathogenic variants may also contribute to the phenomenon in some cases, posing implications for disease transmission risk (Otten et al., 2018). However, human reports have yielded inconsistent findings. For instance, some ESC lines derived from human NT blastocysts containing the same combination of mtDNA haplotypes showed reversion while others did not (Kang et al., 2016; Yamada et al., 2016). Investigations into D-loop polymorphisms revealed inconclusive results in ESC lines (Kang et al., 2016).

Interestingly, the case in Ukraine reported mtDNA reversion in the second child conceived through NT, while the first child did not exhibit genetic drift despite the identical team, oocyte donor, and clinic being involved (Mazur, 2020). Additionally, reports of genetic drift following PGT are intriguing as there is no heterologous mtDNA present (Mitalipov et al., 2014). Overall, this suggests mtDNA sequence-independent mechanisms in mtDNA reversion and instability.

Uneven segregation of mtDNA, possibly influenced by a founder cell effect, may contribute to genetic drift, as seen in mouse and monkey studies (Meirelles and Smith, 1997; Ma et al., 2021). Nevertheless, refinement of NT procedures to minimize carryover and the supposed absence of mtDNA replication during preimplantation development reduce the likelihood of maternal mtDNA propagation solely through random segregation. Another theory proposes the existence of a post-fertilization mtDNA bottleneck, involving the selective replication of a small subpopulation of mtDNA (Wolf et al., 2017). Moreover, the mtDNA replication–transcription balance may influence proliferation in heteroplasmic cells based on cellular bioenergetic demands (Røyrvik and Johnston, 2020).

Mito-nuclear interactions and haplogroup matching

NT introduces a unique scenario, as the transfer of a nucleus into an enucleated oocyte creates a novel combination of nuclear and mitochondrial genomes, potentially leading to mito-nuclear incompatibility. Some researchers have expressed safety concerns related to disrupting mito-nuclear interactions, emphasizing their impact on the fitness, metabolism, and longevity of reconstituted oocytes (Reinhardt et al., 2013; Wolff et al., 2014). However, the arguments behind these concerns have been contested (Chinnery et al., 2014). While some studies in animal models have reported mito-nuclear incompatibility (Holmbeck et al., 2015; Ma et al., 2016), others suggest no harmful effects (Tachibana et al., 2009; Latorre-Pellicer et al., 2016).

Limited reports of experiments investigating mito-nuclear compatibility in humans involved transferring fibroblast nuclei from a patient with Leigh syndrome into recipient oocytes, resulting in pluripotent stem cells with oocyte-derived mtDNA deviating by 47 nucleotide positions. Despite this difference, the cells exhibited transcriptional and metabolic similarities to controls, highlighting adaptability in mito-nuclear interactions in human cells (Ma et al., 2015). Moreover, studies on human NT embryos have shown no adverse effects on mitochondrial function (Kang et al., 2016; Yamada et al., 2016).

Population studies revealed that mtDNA copy number decreases with increasing dissimilarity in ancestry between nDNA and mtDNA among admixed individuals, suggesting suboptimal regulation of mtDNA replication when nDNA and mtDNA genes have different ancestries. However, this trend was only reported in a specific subset of populations (African Americans and Puerto Ricans) (Zaidi and Makova, 2019). Another study analysing heteroplasmic post-mortem tissues described positive selection for reduced mitochondrial function, indicating a degree of tolerance in mito-nuclear compatibility (Rishishwar and Jordan, 2017). Population genetics meta-analyses indicate that deleterious mtDNA and nDNA interactions are unlikely to be more prevalent in individuals born after NT than in normal reproduction (Eyre-Walker, 2017).

Haplogroup matching has been suggested as a possible solution to overcome mito-nuclear incompatibility. Nevertheless, studies showed that even within the same haplogroup, there can be significant differences in mtDNA sequence (Røyrvik et al., 2016; Takeda et al., 2023). In addition, haplogroup matching would substantially limit the number of available donors and potentially increase waiting time for those wishing to access the technique (Takeda et al., 2023).

Mitochondrial gene editing

Based on the success of nuclear gene editing, scientists have sought to adapt current tools for the manipulation of the mitochondrial genome, including the integration of a specific mitochondrial targeting sequence (MTS) into the editing machinery. Therapeutic mitochondrial gene editing aims to reduce the mutation load, resulting in a shift in mtDNA heteroplasmy, improvement in mitochondrial function, and alleviation of clinical symptoms. These technologies can be categorized based on their mechanism of action into nucleases and base editors (Table 6).

While some studies have attempted to use the CRISPR–Cas9 system for mtDNA editing (Jo et al., 2015; Hussain et al., 2021; Bi et al., 2022), its suitability remains controversial due to the difficulties associated with importing guide RNA into the mitochondrial matrix (Gammage et al., 2018a; Schmiderer et al., 2022) and is therefore not discussed in detail in this review.

Nuclease-based strategies

Nuclease-based strategies aim to cleave a specific mtDNA region present in only one haplotype. Since mitochondria lack an efficient DNA repair mechanism, mtDNA is rapidly degraded following double-strand break (DSB) formation (Nissanka et al., 2018; Peeva et al., 2018). Selectively eliminating mutant mtDNA prompts the replication of the residual wild-type molecules, thereby restoring mitochondrial function (Fig. 5A). Mito-nucleases include restriction endonucleases (REs), zinc finger nucleases (ZFNs), transcription activator-like effector nucleases (TALENs), and meganucleases. They are specific for heteroplasmic mtDNA mutations (both point mutations and single

Table 6. Summary of the techniques available to edit the mitochondrial genome and key studies.

Tool		Mechanism of action	DNA binding domain	Targetable sequence	Heteroplasmic point mutations	Heteroplasmic deletions	Homoplasmic mutations	Notes	Key studies
Mito-Nucleases	**Mito-REs**	Elimination of mutant mtDNA following DSB formation	Restriction site dependent	−/+	Yes (limited)	Yes (limited)	No		Srivastava and Moraes (2001), Reddy *et al.* (2015)
	Mito-TALENs	Elimination of mutant mtDNA following DSB formation	TALE repeat	+++	Yes	Yes	No		Bacman *et al.* (2013, 2018), Reddy *et al.* (2015)
	Mito-ZFNs	Elimination of mutant mtDNA following DSB formation	ZF domain	++	Yes	Yes	No		Gammage *et al.* (2014, 2016, 2018b)
	Mito-ARCUS	Elimination of mutant mtDNA following DSB formation	I-Cre based	+++	Yes	Yes	No		Zekonyte *et al.* (2021), Shoop *et al.* (2023b)
	Mito-TevTALE	Elimination of mutant mtDNA following DSB formation	TALE repeat	++	Yes	Yes	No		Pereira *et al.* (2018)
Mito-BEs	**Mito-DdCBEs**	C>T base transition	TALE repeat	+	Yes	No	Yes	Concerns for off-target editing	Mok *et al.* (2020, 2022a), Guo *et al.* (2022), Silva-Pinheiro *et al.* (2023)
	TALEDs	A>G base transition	TALE repeat	+	Yes	No	Yes	Concerns for off-target editing	Cho *et al.* (2022, 2024)

BEs, base editors; REs, restriction endonucleases; TALENs, transcription activator-like effector nucleases; ZFNs, zinc finger nucleases; DdCBEs, DddA-derived cytosine base editors; TALEDs, TALE-linked deaminases.

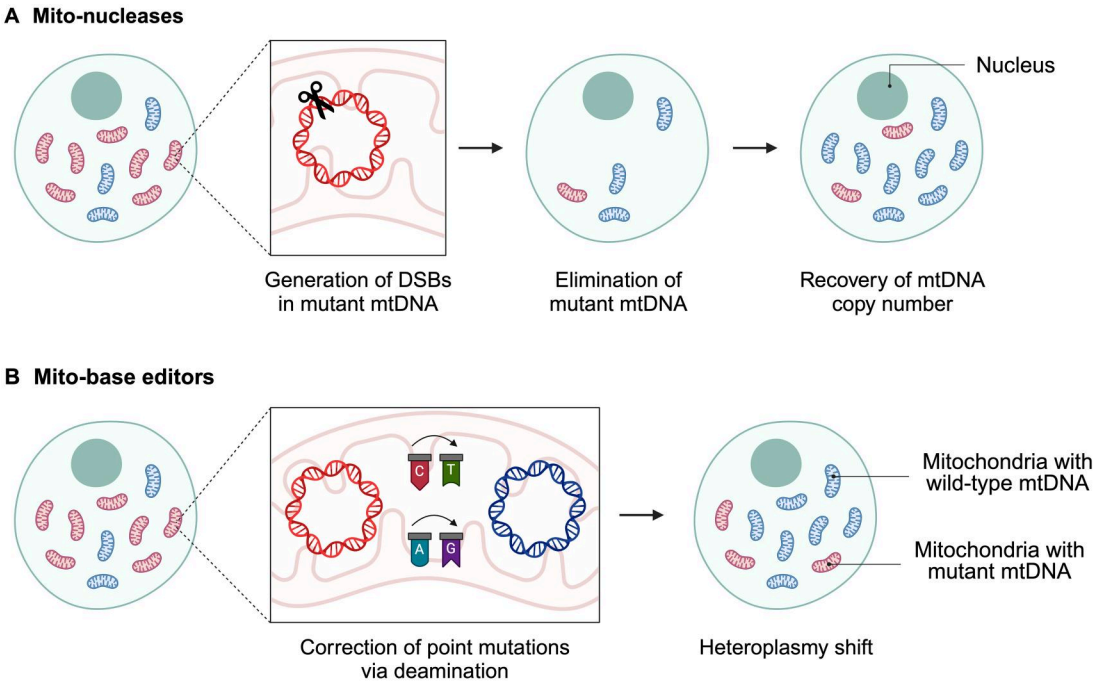


Figure 5. Mitochondrial gene editing approaches. Schematic illustration of the mechanisms of action of genome engineering tools available to edit the mitochondrial DNA (mtDNA). (**A**) Mito-nucleases work by creating double-strand breaks (DSBs) in the mutant mtDNA sequence, which is then degraded. Subsequently, replication of the residual wild-type copies repopulates the mitochondria. (**B**) Mito-base editors generate nucleotide conversion via deamination, inducing a heteroplasmy shift. Figure created with BioRender.

deletions) and have been used in various models, such as cultured cells and embryos, and *in vivo* (Shoop *et al.*, 2023a).

Restriction endonucleases (RE)

This approach has proven successful in eliminating target mtDNA in different models, including mito-PstI in rodent cell lines (Srivastava and Moraes, 2001) and mito-ApaLI in the germline (Reddy *et al.*, 2015) and various tissues *in vivo* of heteroplasmic mice (Bayona-Bafaluy *et al.*, 2005; Bacman *et al.*, 2012). Nevertheless, human applications are limited due to the scarcity of pathogenic mtDNA mutations creating distinct recognition sites. The m.8993T>G mutation remains the only human mtDNA variant efficiently targeted with mito-SmaI (Tanaka *et al.*, 2002) and mito-XmaI (Alexeyev *et al.*, 2008). Cleavage resulted in a significant reduction of mutant mtDNA in patient-derived cybrid cells without compromising cellular health.

Designer nucleases

Designer nucleases such as mito-ZFNs and mito-TALENs have addressed the constraints of mito-REs. These programmable nucleases feature modular, engineerable, sequence-specific DNA-binding domains, and their dimeric architecture enables them to target both mtDNA point mutations and deletions (Table 6).

Mito-ZFNs demonstrated efficacy in reducing mutation load and restoring mitochondrial respiratory function in patient-derived cybrid cells carrying the m.8993T>G mutation and the 'common' 4977 bp deletion (Gammage *et al.*, 2014, 2016). Similarly, mito-TALENs have successfully targeted various pathogenic mtDNA mutations, including m.3243A>G, m.14459G>A, m.8344A>G, and m.13513G>A in cybrid cell lines (Bacman *et al.*, 2013; Hashimoto *et al.*, 2015) and induced pluripotent stem cells (iPSCs) (Yahata *et al.*, 2017; Yang *et al.*, 2018), demonstrating their versatility. Additionally, their application in heteroplasmic oocytes and oocyte–cytoplast fusions showed effective induction of heteroplasmy shift (Reddy *et al.*, 2015; Yang *et al.*, 2018).

As a critical step towards clinical application, both platforms were tested *in vivo* in a heteroplasmic mouse model harbouring the m.5024C>T mutation (Kauppila *et al.*, 2016), equivalent to the pathogenic human m.5650G>A mutation, which causes mitochondrial myopathy (Horváth *et al.*, 2003; McFarland *et al.*, 2008). Studies have shown a specific reduction in mutant mtDNA levels, concomitant with the restoration of physiological function in the targeted tissues (Bacman *et al.*, 2018, 2024; Gammage *et al.*, 2018b).

Their monomeric derivatives, including single-chain mito-ZFN (Minczuk *et al.*, 2008) and mitoTev-TALE (Pereira *et al.*, 2018), effectively reduced mutation load in cybrid cells harbouring the m.8993T>G and m.8344A>G mutations, respectively. Furthermore, the meganuclease, mito-ARCUS, demonstrated efficacy in shifting mtDNA heteroplasmy in mouse and human models (Zekonyte *et al.*, 2021; Shoop *et al.*, 2023b).

Mitochondrially targeted nucleases, while effective for shifting heteroplasmy, cannot address pathological conditions in homoplasmic states. Their inability to introduce specific *de novo* nucleotide changes in mtDNA substantially restricts the creation of meaningful models for mitochondrial disorders. Moreover, mtDNA degradation in oocytes and embryos may lead to mtDNA depletion due to the lack of mtDNA replication during preimplantation development. Consequently, this hinders the compensation for the removal of mutant mtDNA molecules by wild-type ones, posing a potential risk to embryonic development and implantation (Wai *et al.*, 2010; Reddy *et al.*, 2015). As an alternative, mitochondrial base editors have emerged as a promising strategy capable of treating diseases arising from pathogenic mtDNA point mutations in both heteroplasmic and homoplasmic contexts without the risk of mtDNA depletion.

Base editors

A significant breakthrough was made by the development of a novel class of CRISPR-free base editors, which are based on an unconventional bacterial deaminase called double-stranded DNA deaminase toxin A (DddA). This revolutionary technology

enables precise C-to-T editing in mtDNA without inducing DSBs and can be used to correct both heteroplasmic and homoplasmic mutations (Mok *et al.*, 2020) (Fig. 5B).

Most excitingly, applications in various organisms, including human cells (Lee *et al.*, 2023) and embryos (Chen *et al.*, 2022; Wei *et al.*, 2022b), mice (Lee *et al.*, 2021, 2022; Guo *et al.*, 2022; Silva-Pinheiro *et al.*, 2022, 2023), rats (Qi *et al.*, 2021; Tan *et al.*, 2023), and zebrafish (Guo *et al.*, 2021; Sabharwal *et al.*, 2021), have showcased the broad potential of DddA-derived cytosine base editors (DdCBE). Further enhancements, such as zinc finger deaminases (ZFD) (Lim *et al.*, 2022; Willis *et al.*, 2022) and Ddda homologues, like DddA6 and DddA11 (Mok *et al.*, 2022a; Guo *et al.*, 2023; Mi *et al.*, 2023), have improved editing efficiency and expanded the substrate scope. Moreover, a monomeric variant (mDdCBE) has been engineered, increasing versatility and easing delivery with viral vectors (Mok *et al.*, 2022b) (Table 6).

The development of adenine base editors, known as TALE-linked deaminases (TALEDs), further expanded the toolkit for mitochondrial genome editing, enabling A-to-G base conversion (Cho *et al.*, 2022, 2024). This innovation significantly broadened the range of potentially targeted disease-associated mtDNA mutations. Recently, strand-selective base editors were also described for mtDNA editing (Yi *et al.*, 2024) (Table 6).

There are, however, concerns regarding off-target editing (Lei *et al.*, 2022; Wei *et al.*, 2022a) and limitations to targetable mutation types, with current base editors restricted to transition mutagenesis. Nevertheless, the field of mtDNA base editing is advancing rapidly, holding great promise for precision gene editing in mitochondrial disorders. In the coming years, mtDNA editors could similarly be further improved for therapeutic applications.

Emerging research and future directions

Although only a few treatments for mitochondrial disorders, such as Idebenone, have shown efficacy in controlled clinical trials (Carelli *et al.*, 2017), there is ongoing interest in developing novel therapies, driven by progress in understanding mitochondrial biology and detecting genetic abnormalities (Russell *et al.*, 2020; Aldossary *et al.*, 2022).

With advances in sequencing technologies and bioinformatics, the accuracy and speed of mitochondrial disease detection are expected to improve, even at the single-cell level. Integrating multi-omics approaches (including genomics, transcriptomics, proteomics, and metabolomics) will provide a more comprehensive understanding of these disorders and their underlying mechanisms (Wortmann *et al.*, 2022). For example, digital PCR has emerged as an innovative method for the simultaneous quantification of mtDNA heteroplasmy and copy number, greatly increasing the accuracy of analysis (Tytgat *et al.*, 2021; Shoop *et al.*, 2022, 2023a). Moreover, the introduction of NGS may lead to superior outcomes when using PGT, providing highly accurate quantification of mtDNA mutations and simultaneous detection of forms of aneuploidy that affect embryo viability and likelihood of successful IVF (Spath *et al.*, 2021). Long-term follow-up of children born after PGT-MT is also needed, especially when heteroplasmic embryos are transferred, as this could provide further insights into mtDNA segregation and possibly mtDNA reversion.

The crosstalk between the nuclear and mitochondrial genomes is critical for unravelling the pathogenesis of mitochondrial diseases. Future research will increasingly focus on how nuclear gene mutations impact mitochondrial function and how environmental factors, lifestyle choices, and metabolic states influence mitochondrial health and disease progression (Aldossary *et al.*, 2022). This knowledge is key to developing preventive and therapeutic strategies. Additionally, investigating mtDNA segregation and heteroplasmy shifts is pivotal for understanding mitochondrial diseases. These dynamics, shaped by genetic bottlenecks, selective pressures, and quality control mechanisms like mitophagy, are crucial in determining disease outcomes.

Gene editing represents one of the most promising areas of research, particularly for correcting heteroplasmic mtDNA mutations. In the coming years, mtDNA base editors will be likely ready for clinical applications (Moraes, 2024). Future research will also focus on refining delivery methods, minimizing off-target effects, and ensuring overall safety. Additionally, ongoing efforts aim to refine strategies for eliminating residual pathogenic mtDNA in techniques like NT. For instance, inducing mitophagy has shown potential in both mouse and human embryos (Diot *et al.*, 2016; Fan *et al.*, 2022), and combining NT with mitochondrial gene editing offers a promising approach to reducing heteroplasmy (HFEA, 2016; Adashi *et al.*, 2021).

Although the rarity of mitochondrial disorders has historically limited clinical trials, advances in gene therapy, screening assays, and innovative trial designs are expected to drive progress in the coming decade, potentially leading to more personalized treatments.

Legal and ethical considerations

Choosing any of the reproductive options discussed above involves complex considerations, including specific mutations, hereditary risks, legal regulations, technical aspects, and personal preferences.

For instance, PGT-MT significantly reduces the likelihood of an affected foetus and consequently, there is less need to make decisions about pregnancy termination if an affected foetus is detected, but it is more costly than PND-MT. Eligibility for ART is also crucial, with factors such as female age, ovarian reserve, and mtDNA disease manifestation having a powerful influence on PGT-MT outcomes (Verhaak *et al.*, 2007). Conception with donor oocytes presents its own set of issues, including concerns for the donor's welfare, psychological hurdles related to the absence of a genetic link between mother and child, fear of disruption of the family unit in jurisdictions with identifiable donors, and prohibition on the use of gamete donors in some countries.

Patients should be informed about all available options, including those outside fertility clinics, such as choosing not to have children or accepting the associated risks. Comprehensive non-directive counselling of patients is essential, covering not just clinical aspects but also psychological, legal, and ethical ones. For example, when PND is offered, it implies that pregnancy termination will follow if test results indicate the presence of a likely affected foetus. Many countries allow termination for severe diseases, but for mitochondrial disorders, predicting the outcome is particularly challenging. The mutation distribution in foetal organs and tissues must remain stable, and setting cut-off points for mutation load involves balancing the risk of aborting potentially healthy foetuses against providing false reassurance and residual risks for prospective parents. Even with nDNA mutations, phenotypical variability adds to the complexity (Smeets *et al.*, 2023).

Legal restrictions on PGT vary by country, dictating which conditions can be screened for and how embryos are selected (Ginoza and Isasi, 2020). For mtDNA diseases, some parents may

prefer selecting only male embryos to prevent passing residual mutations to future generations. On the other hand, in cases of LHON, which has variable penetrance, selecting female embryos might be considered to minimize disease risk (Black *et al.*, 1995; Yu-Wai-Man *et al.*, 2002). However, legal constraints and the moral implications of embryo destruction, especially with the unpredictable nature of phenotypes, pose additional challenges.

Basic research on NT requires creating embryos specifically for research. However, the Oviedo Convention restricts embryo research to those left over from IVF treatments, prohibiting the creation of embryos solely for research purposes. Since many European countries adhere to this convention, essential NT research is limited to a few countries like Belgium, Sweden, and the UK (Hendriks, 1997). Outside Europe, many countries either forbid this research or restrict federal funding for it (Cohen *et al.*, 2020).

Clinical implementation of NT is regarded by some as a form of germline genome modification, which many countries prohibit (Ishii and Hibino, 2018). This led to the well-known story of the first baby born through MST in the USA, whereby embryo transfer took place in Mexico as the procedure is banned in the USA (Zhang *et al.*, 2017a). The UK has allowed clinical trials involving MST and PNT, but not PBT, issuing a limited number of licenses (clinicaltrials.gov—study NCT04113447) (Cohen *et al.*, 2020), and was criticized for painting an overly optimistic picture of the benefits NT might bring (Herbrand, 2017).

Ethical concerns regarding NT include accessibility issues (e.g. who can afford NT?) (Herbrand, 2017), the suboptimal allocation of scarce resources (Baylis, 2017), inefficient use of donor oocytes, and possible bewilderment for the resulting offspring due to having 'three genetic parents' (Baylis, 2013). However, some argue that the mtDNA donor should not be accounted as a parent (considering that the mitochondrial genome is only ~0.0005% the size of the haploid nuclear genomes contributed by the other parents) (Dimond and Stephens, 2018), and that social responses will impact children's well-being more than the technology itself (Scully, 2017). Moreover, safety concerns related to reversion and mtDNA genetic drift are significant, requiring long-term follow-up of children from clinical trials.

Some suggest that NT trials should focus on patients who do not suffer from mitochondrial disease but might benefit from the procedure to overcome infertility, as reversion in these cases should not have such adverse effects on offspring (Pennings, 2022). However, this does not acknowledge evidence in mice that heteroplasmy involving two 'normal' yet genetically distinct mtDNAs might result in pathology (Lechuga-Vieco *et al.*, 2022). Additionally, this approach conflicts with regulations such as those set by the HFEA, which restrict NT to those at high risk of passing severe mitochondrial diseases (HFEA, 2017). The rationale for this limitation, including the lack of a strong link between infertility and mitochondrial dysfunction, has been questioned (Braun, 2024). An important observation in this debate is that both applications have as their main objective the birth of a healthy, genetically related child. The portrayal of NT as a 'therapy' for mitochondrial diseases (which generates more sympathy than the 'desire' for a child) has been flagged as misleading in this context, as nobody is cured and as the prevention of the birth of a person with a mitochondrial disorder does not require NT (Rulli, 2017). Similar to PGT, NT should be presented as a risk-reduction strategy rather than a guaranteed solution, with ongoing monitoring of resulting children (Ishii and Palacios-González, 2017).

NT might also offer benefits over traditional donor conception by potentially reducing the psychological burden for the donor, who would contribute only small amounts of mtDNA, rather than nDNA (and mtDNA). A recently published study from France confirms this hypothesis (Mayeur *et al.*, 2024), but a survey from Belgium does not, with women indicating equal willingness to donate oocytes for both procedures (Pennings *et al.*, 2024). In the UK, anonymous oocyte donation is allowed for NT, but not for conventional donor conception (Appleby, 2018; Turkmendag, 2018). Some donors may consider NT more controversial, underscoring the importance of clear communication about the use of their donated oocytes (Schaefer, 2018).

Finally, gene editing remains the most contentious and legally challenging option. Many legislators, in line with the Oviedo Convention, prohibit germline modifications, fearing unexpected long-term negative impacts on the descendants and a slippery slope towards eugenic practices (Baylis *et al.*, 2020; Almeida and Ranisch, 2022). However, as far as one avoids 'non-medical' interventions, the ethical debate seems to have shifted from mainly principled objections, such as framing genome editing as an infraction against a natural order, against human dignity, and as disturbing the human gene pool, to mainly pragmatic concerns. These now focus on safety issues, including off-target effects, pleiotropy, and other unforeseen long-term consequences, as well as potential misuse (Almeida and Ranisch, 2022; Segers, 2023). Also within the general public, there seems to be wide societal support for genome editing when it aims to prevent severe diseases (Kubikova *et al.*, 2023), and legislators are increasingly open to the possibility of reversing their stance on this topic (Martin and Turkmendag, 2021). Some philosophers have argued that pursuing germline gene editing is not only morally permissible but even morally desirable (Gyngell *et al.*, 2017). It is important, however, to remain vigilant of the narratives and value judgements about certain human traits that accompany the idea of germline gene editing and that may influence the extent to which people with disabilities feel valued and feel welcome in society (Segers and Mertes, 2020). In addition, the research that would be needed to establish the safety of germline gene editing is fraught with ethical issues, including creating and destroying large numbers of human embryos and the recruitment of oocyte donors (Niemiec and Howard, 2020).

Given the legal and ethical complexities, combined with the numerous uncertainties that patients with mitochondrial diseases face, extensive counselling is necessary to help prospective parents make informed decisions that align with their personal norms and values and comply with local legislation and regulation.

Conclusions

Mitochondrial diseases are a complex group of disorders caused by mutations in both nuclear and mitochondrial genomes, underscoring the crucial role of mitochondria in cellular energy production and overall health. Advances in genetic testing and molecular biology have significantly enhanced our understanding of these diseases, leading to improved diagnostics and the development of novel therapeutic approaches.

Current strategies to prevent the transmission of mtDNA diseases include advanced reproductive techniques, each with unique challenges and ethical considerations. PGT assesses embryos for maternal mtDNA mutations to select those with lower mutation levels. PND, performed during pregnancy, estimates the risk of mtDNA-related diseases at birth, but its accuracy is

hampered by difficulties in measuring mutation stability and heteroplasmy. NT has emerged as a novel procedure, but due to limited evidence, accurately predicting reproductive outcomes remains difficult. Additionally, while genome editing tools targeting mtDNA are promising, they are not yet safe for clinical use.

Continued research into the molecular mechanisms of mitochondrial dysfunction, the development of innovative treatments, and the integration of multidisciplinary approaches will be crucial in addressing the challenges posed by mitochondrial diseases. Collaborative efforts among researchers, clinicians, and patients are essential to drive progress and ultimately improve outcomes for those affected by these complex disorders.

Data availability

There are no new data associated with this article.

Authors' roles

N.C. and B.H. designed the work and drafted 'Mitochondrial Genetics and Function', 'Impacts on Reproduction', 'Germline Nuclear Transfer', and 'Gene Editing'. K.S. drafted 'Reproductive Genetic Testing'. D.L. and J.P. drafted 'Nuclear and Mitochondrial DNA Mutations'. I.F.M.D.C. drafted 'Diagnostic Methods' and 'Current Treatments'. H.M. drafted 'Legal and Ethical Considerations'. D.W. and L.B. contributed to further writing and revision of the manuscript. N.C. produced all tables and figures. All authors provided comments and approved the final version for submission.

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Conflict of interest

The authors declare no conflict of interest regarding the contents of this manuscript.

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